

# **Automated biomass estimation using self-supervised vision transformers**

by

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## SUMMARY

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This research develops a framework that combines self-supervised vision transformer architectures with geometry-grounded 3D reconstruction for automated biomass estimation. The study dives into multiple self-supervised Vision Transformer backbones to perform recognition and structural feature encoding. In addition to that, advanced transformer-based 3D reconstruction models are evaluated to generate plant geometry from RGB-D and multiview images. Multimodal fusion techniques are integrated using 2D semantic and 3D structural features for regression-based biomass prediction. In so doing, the study shall focus on data efficiency and generalizability with validation against ground-truth metrics such as MAE, RMSE, MARE, and Chamfer Distance.

The study contributes a self-supervised, label-efficient pipeline for plant biomass estimation that integrates state-of-the-art transformer models in both 2D representation learning and 3D geometry fusion. This bridges the gap between vision-based phenotyping and plant physiological modeling, advancing the scalability and generalizability of non-invasive biomass prediction tools in precision agriculture and environmental monitoring.

This therefore advances the frontiers of computer vision and machine learning through novel application-driven framework, while simultaneously providing practical tools for agriculture and environmental monitoring. The scientific contributions from improving vision transformer generalization to enabling label-efficient biomass estimation lay a strong foundation for future research at the intersection of artificial intelligence and agricultural engineering. The findings of automated biomass estimation using self-supervised vision transformers are circulated in this dissertation, including additional research journals, articles, and conference proceedings to ensure that the benefits are widely realized across both academia and industry.

## LIST OF ABBREVIATIONS

|              |                                   |
|--------------|-----------------------------------|
| <b>AGB</b>   | above-ground biomass              |
| <b>ANN</b>   | artificial neural network         |
| <b>API</b>   | application programming interface |
| <b>CKA</b>   | centred kernel alignment          |
| <b>CNN</b>   | convolutional neural network      |
| <b>DBH</b>   | diameter at breast height         |
| <b>DINO</b>  | self-distillation with no labels  |
| <b>DL</b>    | deep learning                     |
| <b>DWT</b>   | discrete wavelet transform        |
| <b>EVI</b>   | enhanced vegetation index         |
| <b>FFN</b>   | feed-forward network              |
| <b>GAN</b>   | generative adversarial network    |
| <b>GPU</b>   | graphics processing unit          |
| <b>GRU</b>   | gated recurrent unit              |
| <b>ICP</b>   | iterative closest point           |
| <b>IoU</b>   | intersection over union           |
| <b>LAI</b>   | leaf area index                   |
| <b>LiDAR</b> | light detection and ranging       |
| <b>LOOCV</b> | leave-one-out cross-validation    |
| <b>MAE</b>   | mean absolute error               |
| <b>MHSA</b>  | multi-head self-attention         |
| <b>ML</b>    | machine learning                  |
| <b>MLP</b>   | multilayer perceptron             |
| <b>MLS</b>   | moving least squares              |
| <b>MVS</b>   | multi-view stereo                 |

## **LIST OF ABBREVIATIONS** (continued)

|              |                                        |
|--------------|----------------------------------------|
| <b>NDVI</b>  | normalised difference vegetation index |
| <b>NeRF</b>  | neural radiance field                  |
| <b>PSNR</b>  | peak signal-to-noise ratio             |
| <b>RF</b>    | random forest                          |
| <b>RGB</b>   | red-green-blue                         |
| <b>RGB-D</b> | red-green-blue-depth                   |
| <b>RMSE</b>  | root mean square error                 |
| <b>SAM</b>   | segment anything model                 |
| <b>SfM</b>   | structure from motion                  |
| <b>SOR</b>   | statistical outlier removal            |
| <b>SSL</b>   | self-supervised learning               |
| <b>SVM</b>   | support vector machine                 |
| <b>TLS</b>   | terrestrial laser scanning             |
| <b>ToF</b>   | time-of-flight                         |
| <b>UAV</b>   | unmanned aerial vehicle                |
| <b>VGGT</b>  | visual geometry grounded transformer   |
| <b>ViT</b>   | vision transformer                     |

# LIST OF SYMBOLS

## Image Formation and Sensor

|                     |                                      |
|---------------------|--------------------------------------|
| $u, v$              | pixel column and row coordinates     |
| $D(u, v)$           | depth value at pixel $(u, v)$ (mm)   |
| $W, H_{\text{img}}$ | image width and height (pixels)      |
| $f_x, f_y$          | camera focal lengths (pixels)        |
| $c_x, c_y$          | principal point coordinates (pixels) |
| $\mathbf{P}$        | back-projected 3D point $(X, Y, Z)$  |
| $X, Y, Z$           | 3D Cartesian coordinates (mm)        |

## Point Cloud Registration (ICP)

|                      |                                                      |
|----------------------|------------------------------------------------------|
| $E(\mathbf{T})$      | ICP point-to-plane energy function                   |
| $\mathbf{T}$         | rigid transformation matrix (rotation + translation) |
| $\mathbf{p}_i$       | target point in reference cloud                      |
| $\mathbf{q}_i$       | corresponding source point                           |
| $\hat{\mathbf{n}}_i$ | unit surface normal at $\mathbf{p}_i$                |

## Mesh Geometry and Geometric Features

|                                            |                                               |
|--------------------------------------------|-----------------------------------------------|
| $\mathbf{a}_i, \mathbf{b}_i, \mathbf{c}_i$ | triangle vertex positions                     |
| $A$                                        | mesh surface area ( $\text{m}^2$ )            |
| $V$                                        | mesh enclosed volume ( $\text{m}^3$ )         |
| $H$                                        | plant height – bounding-box $Y$ extent (m)    |
| $V_b$                                      | bounding-box volume ( $\text{m}^3$ )          |
| $\kappa$                                   | compactness = $36\pi V^2/A^3$ (dimensionless) |

### Reconstruction Quality Score

|                 |                                                   |
|-----------------|---------------------------------------------------|
| $Q$             | overall reconstruction quality score $\in [0, 1]$ |
| $Q_{\text{gf}}$ | geometric fidelity sub-score                      |
| $Q_{\text{cb}}$ | coverage balance sub-score                        |
| $Q_{\text{ss}}$ | surface smoothness sub-score                      |
| $Q_{\text{mi}}$ | manifold integrity sub-score                      |

### Neural Network and Representation Learning

|                                      |                                          |
|--------------------------------------|------------------------------------------|
| $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ | query, key, and value matrices in MHSA   |
| $d_k$                                | per-head key dimension in MHSA           |
| $f_{\theta}$                         | neural encoder with parameters $\theta$  |
| $f_{\theta}^s, f_{\theta}^t$         | student and teacher encoders (DINO)      |
| $\mathcal{L}$                        | total training loss                      |
| $\mathcal{L}_{\ell_1}$               | $\ell_1$ photometric reconstruction loss |
| $\mathcal{L}_{\text{GAN}}$           | GAN adversarial loss                     |
| $\lambda_{\ell_1}$                   | weighting coefficient for $\ell_1$ loss  |

### SAM 3D Objects and Gaussian Splat Representation

|                         |                                                                 |
|-------------------------|-----------------------------------------------------------------|
| $\mathcal{O}$           | voxel occupancy grid $\in \{0, 1\}^{64^3}$                      |
| $\mathcal{G}$           | set of 3D Gaussians (splat representation)                      |
| $\boldsymbol{\mu}_i$    | mean position of the $i$ -th Gaussian                           |
| $\boldsymbol{\Sigma}_i$ | covariance matrix of the $i$ -th Gaussian                       |
| $\alpha_i$              | opacity of the $i$ -th Gaussian                                 |
| $\mathbf{c}_i$          | colour of the $i$ -th Gaussian                                  |
| $N_G$                   | total number of Gaussians in splat                              |
| $\sigma_{i,k}$          | $k$ -th principal standard deviation of $\boldsymbol{\Sigma}_i$ |
| $V_G$                   | effective Gaussian volume                                       |
| $Y_{\text{soil}}$       | depth-threshold soil-surface $Y$ coordinate (mm)                |

## Regression and Evaluation

|             |                                                          |
|-------------|----------------------------------------------------------|
| $N$         | number of specimens (plants)                             |
| $y_i$       | measured (ground-truth) dry biomass of specimen $i$ (kg) |
| $\hat{y}_i$ | predicted biomass of specimen $i$ (kg)                   |
| $\bar{y}$   | mean measured biomass across all specimens (kg)          |
| $R^2$       | coefficient of determination                             |



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# CHAPTER 1 INTRODUCTION

## 1.1 BACKGROUND AND MOTIVATION

Above-ground biomass (AGB), the total dry mass of living plant material above the soil surface, is a foundational quantity in ecology, agriculture, and climate science. Accurate AGB estimates enable carbon accounting, assessment of ecosystem productivity, monitoring of tree growth, and data-driven decision making in precision agriculture [2, 3, 4]. For annual crops, biomass tracks physiological vigour and yield trajectory; for ornamental and nursery stock, it is a direct index of plant health and commercial value. The traditional reference method of destructive harvesting, oven drying, and gravimetric weighing is accurate but fundamentally incompatible with repeated monitoring of the same individual and with large-scale phenotyping operations that involve hundreds or thousands of plants [5].

Non-destructive estimation from remote or proximal sensing signals has therefore been an active research frontier for two decades [6, 7, 8]. LiDAR-based approaches, which reconstruct three-dimensional (3-D) point clouds of vegetation canopy, have achieved strong performance in forest and field-crop settings by extracting allometric features such as tree height, crown radius, and canopy volume, then relating these to dry mass via regression or allometric equations [9, 10, 11, 12]. Proximal RGB-D sensing, which combines colour imagery with structured-light or time-of-flight depth, offers a lower-cost and greenhouse-compatible alternative. Nombambela [1] demonstrated that a Microsoft Kinect V2 multi-view scanning protocol with classical geometric feature extraction and Random Forest regression achieves  $R^2 > 0.80$  on pot-grown ornamental saplings, establishing a strong baseline for the sensor modality addressed in this dissertation.

Despite this progress, two unresolved problems constrain the accuracy of RGB-D-based biomass pipelines. The first is the *pot contamination problem*: when depth-based 3-D reconstruction is per-



formed without explicit plant-to-substrate segmentation, the reconstructed mesh includes the planting pot and soil surface, introducing a systematic positive volume bias that is both plant-dependent (different species produce different soil exposure patterns) and pot-size-dependent. The second is the *representational bottleneck*: classical pipelines compress the full three-dimensional morphology of a reconstructed plant into a small set of hand-crafted scalar features (volume, surface area, height, base volume, and derived ratios), discarding potentially discriminative textural, topological, and branching information that a sufficiently expressive learned representation could exploit.

Concurrently, the computer vision community has undergone a paradigm shift driven by self-supervised learning and Vision Transformers (ViTs) [13, 14, 15]. Self-supervised methods, including contrastive approaches such as SimCLR [16] and self-distillation methods such as DINO [17], enable rich visual representations to be learned from large unlabelled image collections without task-specific annotation. DINOv2 [18], the most recent iteration of this paradigm, pre-trains a ViT-B/14 encoder on 142 million curated images (LVD-142M) using a combined teacher–student distillation and masked-image-modelling objective. DINOv2 features have been shown to transfer remarkably well to tasks for which the pre-training data contained no explicit examples, including monocular depth estimation, semantic segmentation, and surface normal prediction [18, 19].

Meta AI’s SAM 3D Objects [20], introduced in November 2025, integrates DINOv2 as its image encoder within a feed-forward geometry estimation architecture that produces a voxel-and-mesh 3-D reconstruction from a single RGB image in approximately 3 seconds on consumer hardware. This model provides a natural vehicle for operationalising DINOv2 self-supervised features in a 3-D biomass estimation pipeline without requiring depth sensors or calibrated multi-view rigs at inference time.

The intersection of these developments, namely the pot contamination problem, the representational bottleneck, and the emergence of large-scale self-supervised ViT-based 3-D reconstruction, defines the research space addressed by this dissertation. Following the first-principles three-R framework of Malik et al. (recognition, reconstruction, reorganisation) [21], the proposed *BioViT3R* pipeline integrates a depth-threshold above-ground mask with SAM 3D Objects to simultaneously mitigate pot contamination and replace hand-crafted geometric features with DINOv2’s rich, pre-trained visual representations.

## 1.2 PROBLEM STATEMENT

Accurate, non-destructive, and scalable estimation of above-ground plant biomass from RGB-D imagery is limited by two co-occurring problems that prior work has not jointly addressed.

**Problem 1: Pot Contamination.** Classical depth-based pipelines reconstruct a fused mesh from multiple depth frames and extract volumetric features from the entire visible structure. When the plant is pot-grown, the pot and soil substrate contribute a substantial and variable fraction of the reconstructed volume. On the 40-plant dataset used in this study, the raw classical pipeline overestimates above-ground volume by a mean of 18.4 % relative to manually annotated above-ground extent. Because pot size is not a proxy for plant biomass, this contamination introduces noise into the feature–biomass regression and suppresses the effective  $R^2$  achievable with classical feature sets.

**Problem 2: Representational Bottleneck.** Even when pot contamination is controlled, the six geometric features extracted from the mesh (volume  $V$ , surface area  $A$ , height  $H$ , base volume  $V_b$ , area-to-volume ratio  $A/V$ , and height-to-volume ratio  $H/V$ ) constitute a lossy compression of morphological information. Plants with similar scalar descriptors but different branching architecture, leaf density, or surface texture can have substantially different dry biomass values. The ANN regressor trained on these features reaches a ceiling at  $R^2 = 0.752$  despite hyperparameter optimisation and the addition of a seventh quality-weighted composite feature, confirming that the bottleneck is in the feature representation rather than the regression model.

This dissertation addresses both problems simultaneously through (i) a lightweight depth-threshold mask that isolates above-ground plant structure from the planting pot prior to volume estimation, and (ii) integration of SAM 3D Objects [20], whose DINOv2 [18] image encoder provides self-supervised ViT features that are substantially richer than hand-crafted geometric scalars.

## 1.3 RESEARCH OBJECTIVES AND QUESTIONS

The overarching objective of this research is to investigate whether self-supervised Vision Transformer features, operationalised through a foundation-model 3-D reconstruction pipeline, provide statistically significant improvements over classical and neural-voxel baselines for above-ground biomass estimation of pot-grown ornamental tree saplings captured with a Microsoft Kinect V2 RGB-D sensor. In order to meet this objective, the following specific objectives must be accomplished:

1. To conduct a comprehensive literature study of above-ground biomass estimation methods, 3-D reconstruction techniques, and self-supervised Vision Transformer architectures, so as to analyse their strengths, weaknesses, and applicability to the plant phenotyping domain.
2. To design and collect a novel multi-view RGB-D dataset of pot-grown ornamental tree saplings using a 30-degree angular dual-camera Kinect V2 protocol, with paired destructive dry-biomass ground truth.
3. To implement and evaluate a classical depth-based 3-D reconstruction and geometric feature regression pipeline as a reproducible Level-1 baseline.
4. To design and evaluate a depth-threshold above-ground segmentation mask that isolates plant structure from the planting pot and soil substrate, and to quantify its contribution to biomass estimation accuracy.
5. To integrate the SAM 3D Objects foundation model, whose image encoder is the DINOv2 self-supervised Vision Transformer, as a Level-3 biomass estimation pipeline, and to compare its performance against the classical baseline and a DeepVoxels neural-voxel Level-2 baseline under Leave-One-Out Cross-Validation.

In addressing these objectives, the following four research questions guide the investigation:

- RQ1.** Can classical depth-based 3-D reconstruction provide accurate above-ground biomass estimates for pot-grown ornamental saplings, and what is the upper bound of accuracy achievable with hand-crafted geometric features?
- RQ2.** Do learned 3-D voxel representations (DeepVoxels) improve upon classical geometric features for biomass regression, and if so, under what conditions?
- RQ3.** Do self-supervised ViT features, obtained through the DINOv2 encoder embedded in SAM 3D Objects, provide statistically significant and practically meaningful improvements in biomass estimation accuracy compared with classical and neural-voxel baselines?
- RQ4.** Does increasing the angular density of the multi-view scanning protocol from 90° to 30° intervals improve 3-D reconstruction quality and downstream biomass estimation accuracy?

#### 1.4 SCOPE AND LIMITATIONS OF THE STUDY

This dissertation focuses on pot-grown ornamental tree saplings measured with Microsoft Kinect V2 RGB-D cameras in a controlled indoor environment. Ground-truth biomass is obtained by destructive dry-weight measurement. The experimental dataset comprises 40 individual plants, and all model

comparisons are performed using Leave-One-Out Cross-Validation (LOOCV) to maximise statistical efficiency given the dataset size.

The study is bounded in the following ways. First, results are reported for a single sensor modality (Kinect V2 structured-light RGB-D); generalisation to LiDAR, stereo cameras, or UAV RGB imagery is not evaluated. Second, species diversity within the 40-plant cohort is limited to the ornamental saplings available for the study period; results may not generalise without additional data collection across species and growth stages. Third, the SAM 3D Objects model is used in zero-shot inference mode (no domain-specific fine-tuning), so the 3-D reconstruction accuracy on highly anisotropic plant geometries is constrained by the model's Objaverse-XL pre-training distribution [20, 22]. Fourth, experimental results reported in Chapter 4 are interim and reflect the current state of the ongoing experiments; final results will be updated prior to dissertation submission.

## 1.5 APPROACH

The research follows a three-level methodological hierarchy designed to progressively address the two central problems identified above. At Level 1, a classical depth-based pipeline extending the prior work of Nombambela [1] is implemented on the Kinect V2 sensor: multi-view point clouds are registered via Iterative Closest Point (ICP), meshed using Marching Cubes, and six geometric features are extracted for Random Forest and Artificial Neural Network regression under Leave-One-Out Cross-Validation. A lightweight depth-threshold above-ground segmentation mask is introduced at this level to isolate the plant from pot and soil. At Level 2, the DeepVoxels neural architecture [23] replaces hand-crafted geometric features with a learned  $32^3$  voxel embedding, evaluated as a task-agnostic neural baseline. At Level 3, Meta AI's SAM 3D Objects foundation model [20], whose image encoder is DINOv2 ViT-B/14 pre-trained self-supervisedly on 142 million images [18], is integrated into the pipeline to provide rich visual representations for 3-D reconstruction and biomass regression. Performance is assessed using the same LOOCV protocol across all three levels, with Wilcoxon signed-rank tests applied to validate statistical significance of differences.

## 1.6 RESEARCH CONTRIBUTION

This dissertation makes the following five original contributions.

1. **Novel 30-Degree Angular RGB-D Dataset.** A 40-plant multi-view RGB-D dataset of ornamental tree saplings captured at  $30^\circ$  angular intervals using two synchronised Kinect V2 cameras, paired with destructive dry-biomass ground truth. The acquisition protocol uses an Arduino-

automated turntable and a Jetson Nano synchronisation controller, providing a reproducible, low-cost procedure documented in full for replication.

2. **Depth-Based Above-Ground Segmentation Mask.** A lightweight, training-free depth-threshold method that isolates above-ground plant structure from planting pot and soil. The mask achieves mean IoU =  $0.847 \pm 0.061$  against manual annotations and reduces classical pipeline RMSE by 10.6 % without GPU inference or labelled segmentation data.
3. **First Application of DINOv2 Self-Supervised ViT to 3-D Plant Biomass Estimation.** Integration of SAM 3D Objects, whose image encoder is DINOv2 ViT-B/14 pre-trained on 142 million unlabelled images [18], into a plant biomass estimation pipeline. This constitutes the first reported use of self-supervised Vision Transformer features for 3-D plant reconstruction and regression-based biomass prediction.
4. **Pot Contamination Decomposition Framework.** A systematic ablation that decomposes the total accuracy gain into independent contributions from pot isolation (10.6 % RMSE reduction, RMSE  $0.264 \rightarrow 0.236$  kg) and ViT representation quality (9.7 % RMSE reduction, RMSE  $0.236 \rightarrow 0.213$  kg), providing a principled methodology replicable in future RGB-D biomass research.
5. **Three-Level Comparative Benchmark.** A unified LOOCV benchmark comparing classical geometric regression (Level 1), neural voxel embedding via DeepVoxels [23] (Level 2), and the proposed self-supervised ViT-based method via SAM 3D Objects [20] (Level 3), all evaluated on the same 40-plant dataset with Wilcoxon signed-rank statistical significance testing.

## 1.7 RESEARCH OUTPUTS

The following peer-reviewed research outputs have been produced in the course of this study.

### 1.7.1 Journal Articles

1. A. Masuba, H. C. Myburgh, and A. De Freitas, “Self-Supervised Vision Transformers for Semantic Reconstruction and Biomass Estimation from Multiview Plant Imagery,” *International Journal of Computer Vision* (submitted, 2025) [24].
2. A. Masuba, H. C. Myburgh, and A. De Freitas, “Representation Learning and Geometry Fusion for Generative 3D Reconstruction and Estimation,” *International Journal of Computer Vision* (work in progress) [25].
3. A. Masuba and F. Khan, “Foundation VolumetricViT for 3D Plant Reconstruction and Biomass Estimation,” (work in progress) [26].

### 1.7.2 Conference Articles

1. A. Masuba, H. C. Myburgh, and A. De Freitas, “BioViT3R: Self-Supervised Vision Transformers for 3D Biomass Estimation,” in *Proceedings of the Southern Africa Telecommunication Networks and Applications Conference (SATNAC)*, 2025 [27].

## 1.8 THESIS OVERVIEW

The remaining chapters of this dissertation are organised as follows. In Chapter 2, a comprehensive literature study of above-ground biomass estimation methods, 3-D reconstruction techniques (classical and neural), self-supervised representation learning, Vision Transformers, and foundation models for visual perception is presented. A gap analysis at the end of the chapter identifies the specific limitations that the proposed approach targets and contextualises the novelty of the contributions. In Chapter 3, the research methodology is described in detail. The dataset collection procedure, Kinect V2 sensor characteristics, 30-degree angular protocol, three-level methodological hierarchy, depth-threshold masking algorithm, and evaluation framework are all presented. In Chapter 4, the experimental results are presented. Reconstruction quality, geometric feature statistics, regression performance, ablation studies, pot contamination quantification, and cross-method comparison for all three levels, evaluated under LOOCV on the 40-plant dataset, are reported. In Chapter 5, the results are discussed in the context of the four research questions. The mechanisms behind the observed accuracy improvements are analysed, the findings are situated in the broader literature, and the principal limitations of the study are identified. In Chapter 6, the conclusions and recommendations are presented. The research contributions are summarised, each research question is answered, and six directions for future work are proposed, including domain-specific fine-tuning of SAM 3D Objects, multi-modal sensor fusion, temporal biomass monitoring, and integration with precision agriculture platforms.

## **CHAPTER 2 LITERATURE STUDY**

### **2.1 CHAPTER OVERVIEW**

In this chapter, a comprehensive review of the body of knowledge underpinning this research is presented. The goal is to establish what is currently known about above-ground biomass estimation from proximal and remote sensing, 3-D reconstruction, and self-supervised visual representation learning, and to identify the specific gaps that the proposed approach targets. The rest of this chapter is organised as follows. In Section 2.2, methods for above-ground biomass estimation are surveyed, spanning destructive reference methods, LiDAR-based remote sensing, spectral vegetation indices, and proximal RGB-D sensing. In Section 2.3, 3-D reconstruction techniques are examined, from classical iterative point-cloud registration through to neural implicit and feed-forward architectures. In Section 2.4, self-supervised representation learning is covered, with emphasis on the contrastive and self-distillation methods most relevant to ViT pre-training. In Section 2.5, Vision Transformer architectures and their distinguishing properties relative to convolutional baselines are detailed. In Section 2.6, foundation models for visual perception, including SAM, SAM 3D Objects, and VGGT, are surveyed. In Section 2.7, the application of deep learning to plant phenotyping and biomass estimation is reviewed. Conclusive remarks of this chapter are provided in Section 2.8, which synthesises a gap analysis that motivates the proposed approach.

### **2.2 ABOVE-GROUND BIOMASS ESTIMATION**

#### **2.2.1 Destructive and Allometric Methods**

The gold standard for AGB measurement remains destructive harvesting: the plant is separated into structural components, oven-dried at 60–105 °C to constant mass, and weighed [5]. While accurate, destructive sampling precludes longitudinal monitoring of individuals, demands significant labour, and is impractical at the scale required for modern phenotyping pipelines.

Allometric equations relate easily-measured surrogates, namely trunk diameter at breast height (DBH), height, or crown diameter, to total biomass through empirical power-law relationships calibrated on species-specific or regional datasets [3, 2]. These equations offer a practical non-destructive alternative for forest inventory but suffer from poor transferability across species, sites, and stand conditions [4]. Species-level calibration is resource-intensive, and allometric relationships break down for juvenile plants, multi-stemmed shrubs, and horticultural stock where trunk DBH is either absent or a poor proxy for total mass [5].

### 2.2.2 Remote Sensing and LiDAR-Based Estimation

LiDAR (Light Detection and Ranging) has become the most widely adopted technology for large-scale, non-destructive AGB estimation in both forest and agricultural contexts [9, 28, 8, 10]. Airborne LiDAR systems reconstruct canopy height models and compute volume integrals from which biomass is inferred via allometric or machine-learning models [9, 29, 30]. Terrestrial LiDAR scanning at the plot or individual-tree level achieves higher resolution and enables direct measurement of trunk form, branch architecture, and leaf-area index [12, 31]. Wijaya et al. [28] combined UAV-mounted LiDAR with photogrammetric point clouds to estimate biomass in tall-grass savannas, achieving RMSE below 15% of mean biomass. Pan et al. [8] demonstrated that LiDAR-derived crown metrics fed to a Random Forest regressor outperform simple allometric equations on heterogeneous tropical species.

Jung et al. [32] demonstrated that single-scan terrestrial laser scanning combined with a projection-based deep learning approach can estimate individual tree biomass on par with multi-scan methods, substantially reducing acquisition time and complexity, a finding directly relevant to the single-scan philosophy adopted in the proposed Level-3 method.

Despite these advances, LiDAR systems incur high capital cost, require calibrated sensor-to-platform integration, and remain difficult to deploy in enclosed greenhouse environments where multi-path reflections and occlusion from neighbouring plants degrade point-cloud quality [33]. Passive RGB imaging and active RGB-D sensing offer cheaper and more flexible alternatives for controlled-environment agriculture.

### 2.2.3 Vegetation Indices and 2-D Image-Based Estimation

Spectral vegetation indices derived from multispectral or hyperspectral imagery, notably NDVI, EVI, and LAI, have been used as biomass proxies since the 1970s [6]. Machine learning models including Support Vector Machines (SVM) and Random Forest have been applied to large multi-spectral remote



sensing datasets to map crop biomass at field scale [34, 35, 36]. However, 2-D spectral indices project a 3-D volume onto a plane, discarding the height and structural information that are strongly correlated with dry mass [7, 33].

Boogaard et al. [7] used overhead RGB and multispectral cameras on a field phenotyping platform and applied a CNN to predict wheat biomass, achieving  $R^2 = 0.81$ . Schunck et al. [37] combined RGB and near-infrared imagery with explicit 3-D point cloud features extracted from a UAV platform, reporting  $R^2 \approx 0.78$  for sorghum AGB. Both studies demonstrate that incorporating 3-D geometric cues improves upon purely spectral approaches, but remain dependent on crop-specific training data.

#### 2.2.4 RGB-D Proximal Sensing

Microsoft Kinect and Intel RealSense sensors provide synchronised colour and depth at low cost (< USD 200), making them attractive for greenhouse and nursery phenotyping. Ma et al. [38] used an OAK-D stereo-vision depth camera to estimate biomass of pot-grown plants by extracting volume and surface area features from depth-registered point clouds, achieving  $R^2 = 0.79$  on lettuce and spinach under controlled illumination. Buxbaum et al. [39] deployed proximal RGB-D cameras above potted plants and trained an end-to-end convolutional network directly on depth imagery to predict daily biomass increments with high spatio-temporal resolution, demonstrating that low-cost RGB-D sensors are suitable for longitudinal non-destructive growth monitoring. Cardenas-Gallegos et al. [40] combined RGB-D imaging with morphometric descriptors and machine learning regression to estimate biomass in hydroponic lettuce, achieving competitive accuracy and confirming that RGB-D-derived geometric descriptors generalise across hydroponic and pot-grown controlled-environment contexts. Alabbasi et al. [41] demonstrated early feasibility of Kinect-based plant volume estimation but were limited by the sensor's operating range and susceptibility to infrared interference. Nombambela [1] established the direct precedent for this dissertation: a four-view 90° Kinect V2 scanning protocol with PassThrough filtering, Statistical Outlier Removal, Moving Least Squares surface smoothing, and ICP registration achieved  $R^2 > 0.80$  for pot-grown ornamental saplings. The present work extends Nombambela's protocol to 30° angular intervals with dual cameras, and introduces depth-threshold above-ground masking and a self-supervised ViT reconstruction stage.

### 2.3 THREE-DIMENSIONAL RECONSTRUCTION

#### 2.3.1 Classical Point-Cloud Processing

Classical 3-D reconstruction from range data involves four principal stages: filtering, registration, surface reconstruction, and meshing. Statistical Outlier Removal (SOR) eliminates points whose mean

distance to their  $k$ -nearest neighbours exceeds  $n_\sigma$  standard deviations above the dataset mean [42]. Moving Least Squares (MLS) smoothing fits a local polynomial surface in a neighbourhood radius and re-samples the point cloud onto that surface, suppressing high-frequency sensor noise while preserving macroscopic geometry.

Iterative Closest Point (ICP) [43] registers pairs of partially overlapping point clouds by alternately finding closest-point correspondences and computing the rigid transformation (rotation  $\mathbf{R}$ , translation  $\mathbf{t}$ ) that minimises mean squared correspondence distance. ICP converges to a local minimum; wide-baseline views require coarse alignment initialisation, typically through turntable encoder feedback or feature-point matching [42].

Marching Cubes [44] converts a volumetric implicit representation (voxel grid) into an explicit triangular mesh by classifying each voxel as inside or outside the surface and extracting the intersection iso-surface. The resulting mesh supports direct computation of volume (via the divergence theorem) and surface area (via triangle face norms). Wu et al. [45] applied this classical pipeline to multi-view Kinect images of fruit trees, extracting crown volume features for yield prediction.

### 2.3.2 Structure from Motion and Multi-View Stereo

Structure from Motion (SfM) jointly estimates camera poses and a sparse 3-D point cloud from an unordered image collection by detecting and matching local feature descriptors [42]. Multi-View Stereo (MVS) densifies the SfM sparse cloud by photometric consistency matching [46]. SfM–MVS pipelines are sensor-agnostic (requiring only RGB images) and have been used extensively in field phenotyping [37, 45]. However, their accuracy degrades under textureless surfaces (bare stems), specular reflections (wet leaves), and wide inter-frame baselines, all of which are common in plant scanning scenarios.

### 2.3.3 Neural Implicit and Feed-Forward 3-D Reconstruction

Neural Radiance Fields (NeRF) [47] model a scene as a continuous volumetric radiance function  $f(\mathbf{x}, \mathbf{d}) \rightarrow (\mathbf{c}, \sigma)$  parameterised by an MLP, rendered differentiably via volumetric ray marching. NeRF achieves photorealistic novel-view synthesis but requires per-scene optimisation taking hours, making it impractical for high-throughput phenotyping.

DeepVoxels [23] pre-trains a learnable 3-D voxel grid ( $32^3$  voxels, 64 channels) per scene using a convolutional encoder–decoder with GAN discriminator and  $\ell_1$  photometric loss. The resulting

voxel embedding encodes both geometry and appearance in a compact latent space without explicit point-cloud or mesh representations. Sitzmann et al. demonstrated generalisation to novel viewpoints on synthetic objects; in this dissertation, DeepVoxels embeddings are used as Level-2 feature vectors for biomass regression, providing a learned alternative to classical geometric features without requiring a 3-D mesh.

Bornand et al. [48] addressed the complementary problem of 3-D point cloud completion for individual trees: a deep learning architecture trained on terrestrial LiDAR data predicts missing canopy geometry caused by occlusion in single-scan protocols, recovering internal branch architecture that single-viewpoint acquisition cannot observe. This work demonstrates that learned priors over tree geometry can compensate for occlusion-induced incompleteness, a challenge directly analogous to the canopy-void estimation problem faced by SAM 3D Objects when reconstructing plant canopies from a single image.

DUS3R [49] introduced the concept of feed-forward 3-D reconstruction: given two or more images as input, a transformer-based network directly regresses dense 3-D point maps in a shared coordinate frame without requiring camera calibration or iterative optimisation. MAS3R [50] extended DUS3R with discriminative matching features for wider-baseline generalisation. VGGT (Visual Geometry Grounded Transformer) [51], developed jointly by the Oxford Visual Geometry Group and Meta AI and presented at CVPR 2025, pushed this paradigm further: a single forward pass over multi-view images directly predicts camera intrinsics and extrinsics, depth maps, 3-D point clouds, and surface normals in under 0.2 seconds. Masuba [52] evaluated VGGT as a potential reconstruction backbone for plant biomass estimation, noting that VGGT’s implicit geometric inductive biases and feed-forward inference offer complementary advantages to SAM 3D Objects’ single-image capability.

SAM 3D Objects [20], introduced by Meta AI in November 2025, integrates a DINOv2 ViT-B/14 image encoder with a Geometry Model that predicts a voxel grid and watertight mesh from a single RGB image. The model is pre-trained on Objaverse-XL [22], a dataset of over 10 million 3-D object assets. SAM 3D Objects requires no depth input at inference, processes a single image in approximately 3.2 seconds on a consumer GPU, and outputs a dense triangular mesh from which volume and surface area can be computed directly.

## 2.4 SELF-SUPERVISED REPRESENTATION LEARNING

### 2.4.1 Contrastive and Predictive Pre-Training

Self-supervised learning (SSL) trains a neural encoder  $f_\theta$  to produce representations that are useful for downstream tasks without requiring ground-truth labels during pre-training [14, 53]. The core intuition is that a model forced to reconstruct corrupted or masked views of the same underlying data will learn the low-dimensional generative factors relevant to that data.

SimCLR [16] operationalises this via contrastive loss: two augmented views of the same image (positive pair) are mapped to an embedding space where their distance is minimised, while distances to embeddings of different images (negative pairs) are maximised. BiT (Big Transfer) [54] demonstrated that large-scale supervised pre-training followed by careful fine-tuning transfer yields strong results across diverse downstream tasks, establishing the transfer learning paradigm that SSL methods later outperformed. S4L (Self-Supervised Semi-Supervised Learning) [55] explored hybrid approaches combining a small labelled set with self-supervised pre-training on unlabelled data, directly relevant to the label-efficient setting targeted by this dissertation.

### 2.4.2 Self-Distillation Methods: DINO and DINOv2

DINO (Self-Distillation with No Labels) [17] introduced a teacher–student training architecture in which a student network  $f_\theta^s$  is trained to predict the softmax output of an exponential moving-average (EMA) teacher  $f_\theta^t$  on a set of multi-crop augmented views of each image. A centering and sharpening mechanism in the teacher output prevents representation collapse. Caron et al. demonstrated that DINO-trained ViT features exhibit emergent semantic segmentation properties: the [CLS] token attends to semantically coherent object regions without any segmentation supervision, suggesting that the representations capture structural scene parsing implicitly.

SiT (Self-supervised Image Transformer) [56] extended DINO by incorporating masked-patch prediction alongside multi-crop distillation, enabling the encoder to capture both local patch statistics and global structural context simultaneously.

DINOv2 [18] scaled the DINO paradigm with three key changes: (i) a curated 142-million-image dataset (LVD-142M) constructed via retrieval from internet data; (ii) a combined training objective that interleaves multi-crop distillation with masked-image modelling [56]; and (iii) regularisation via online KNN probing of a held-out evaluation set. The resulting ViT-B/14 checkpoint achieves state-of-the-art

linear-probe accuracy on ImageNet-1k and, critically, demonstrates strong zero-shot generalisation to depth estimation, semantic segmentation, and surface reconstruction tasks, all of which are directly relevant to plant phenotyping [18].

Darcet et al. [19] identified and resolved an artefact in standard DINOv2 features: low-informative background tokens in large ViT models carry inflated attention weights, causing spurious high-frequency patterns in attention maps. Introducing a fixed number of *register tokens*, which are additional learnable embeddings that absorb global scene statistics and thereby free the background patch tokens, eliminates this artefact without degrading classification accuracy and significantly improves the spatial resolution of DINOv2 attention maps.

### 2.4.3 Other Self-Supervised Approaches

Masked image modelling (MAE-style) methods, typified by BEiT and MAE, mask random subsets of input patches and train the encoder to reconstruct them from the unmasked context, yielding representations strong in local texture and global structure prediction [18]. BigSSL [57] demonstrated that self-supervised objectives can scale to billion-parameter models on speech and vision data with consistent downstream gains. Azizi et al. [58] showed that self-supervised pre-training on medical images, even without labels, produces features more transferable to downstream medical classification tasks than ImageNet-supervised baselines, a result directly analogous to the plant domain where supervised pre-training data is scarce.

## 2.5 VISION TRANSFORMERS

### 2.5.1 Architecture

The Vision Transformer (ViT) [13] adapts the Transformer architecture [15] to image data by partitioning an  $H \times W$  image into a grid of  $N = HW/P^2$  non-overlapping  $P \times P$  patches, projecting each patch to a  $D$ -dimensional embedding via a linear layer, prepending a learnable [CLS] classification token, and adding learnable 1-D positional embeddings [13]. The resulting sequence of  $N + 1$  tokens is processed by  $L$  stacked Transformer encoder layers, each comprising Multi-Head Self-Attention (MHSA) and a position-wise feed-forward network (FFN), with layer normalisation and residual connections. The [CLS] token at the final layer aggregates global image information and is used as the image representation for downstream tasks.

MHSA computes attention as:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (2.1)$$

where  $\mathbf{Q}$ ,  $\mathbf{K}$ , and  $\mathbf{V}$  are query, key, and value projections of the input token sequence and  $d_k$  is the per-head dimension. The quadratic complexity of attention in  $N$  enables every patch to directly attend to every other patch, providing global receptive fields from the first layer, a property that CNN architectures achieve only in deeper layers through local receptive-field stacking.

Compared with CNNs, ViTs exhibit weaker inductive biases (no built-in translation equivariance or local connectivity), requiring larger pre-training datasets to match CNN performance [13, 14]. However, when sufficient data is available, ViTs demonstrate superior performance on tasks requiring global spatial reasoning [13] and transfer more effectively across heterogeneous domains [18, 53].

### 2.5.2 ViT Variants Relevant to Biomass Estimation

Simonyan and Zisserman's VGGNet [59] established deep convolutional feature extraction as the standard backbone for visual recognition prior to the ViT era. Kornblith et al. [60] demonstrated through centred kernel alignment (CKA) that features from different architectures, including early ViTs, encode semantically similar representations in spite of architectural differences, suggesting that representation quality is a more fundamental determinant of downstream performance than architectural choice.

Zhang and Liu [61] applied ViT-based feature extraction to hyperspectral wheat canopy imagery for agronomic trait prediction, reporting 12% RMSE reduction over CNN baselines and highlighting ViTs' ability to capture long-range spectral-spatial dependencies. Hanan et al. [62] used ViT backbone features fused with point-cloud descriptors for crop canopy biomass estimation from UAV RGB-D imagery, achieving competitive performance with lower labelled data requirements than CNN-based alternatives. Hindel et al. [63] proposed BalViT, a balanced Vision Transformer training scheme for few-shot remote sensing classification, demonstrating that ViT representations are particularly data-efficient when properly regularised.

### 2.5.3 ViTs for 3-D Understanding

ViTs have been extended to 3-D scene understanding through several mechanisms. Shi et al. [64] proposed 3-D attention grounding, augmenting ViT patch tokens with 3-D positional encodings derived from metric depth maps to produce spatially grounded features. Yin et al. [65] introduced geometry-grounded attention, conditioning the attention computation on epipolar geometry constraints

for multi-view consistency. Qi et al. [66] developed ReCon, a multi-modal ViT that jointly learns 2-D visual and 3-D point-cloud representations via cross-modal distillation. An et al. [67] demonstrated few-shot 3-D object understanding using ViT features extracted from rendered views, relevant to the single-image 3-D reconstruction setting used in this dissertation.

## **2.6 FOUNDATION MODELS FOR VISUAL PERCEPTION**

### **2.6.1 Segment Anything Model**

The Segment Anything Model (SAM) [68], released by Meta AI in 2023, introduced the concept of promptable segmentation: given a point, bounding box, or text prompt, SAM segments the corresponding object or region at interactive speed. SAM is pre-trained on SA-1B, a dataset of over one billion masks crowdsourced from 11 million images. Its image encoder is a ViT-H (632M parameters) pre-trained with MAE, and its prompt encoder and mask decoder are lightweight MLP modules.

SAM established the foundation model paradigm for visual perception: a single large model pre-trained on massive, diverse data generalises to novel tasks and domains in zero-shot or few-shot regimes [68]. Its successor, Grounding DINO [69], extended promptability to open-vocabulary text by fusing a DINO text encoder with SAM's image encoder, enabling arbitrary-class object detection without task-specific fine-tuning.

### **2.6.2 SAM 3D Objects**

SAM 3D Objects [20] extends the SAM paradigm to 3-D by replacing the 2-D mask decoder with a Geometry Model that predicts a voxel grid and watertight mesh from a single RGB image. Critically, the image encoder in SAM 3D Objects is DINOv2 ViT-B/14 [18], exploiting the rich 3-D-aware features demonstrated by DINOv2 in depth estimation benchmarks. The model is pre-trained on Objaverse-XL [22], which provides diverse 3-D assets covering furniture, vehicles, tools, and organic shapes including some botanical objects.

SAM 3D Objects produces three output representations: a  $128^3$  voxel occupancy grid, a watertight triangular mesh, and a Gaussian splat suitable for real-time rendering. From the mesh, volume and surface area are computed via standard geometric algorithms. Because SAM 3D Objects requires only a single RGB image and performs inference in approximately 3.2 seconds, it is well-suited to greenhouse-scale phenotyping where multi-view turntable acquisition is time-consuming.

The principal limitation of SAM 3D Objects for plant biomass estimation is its Objaverse-XL pre-training distribution: the training data contains relatively few highly anisotropic branching structures resembling plants, and the solid-object prior encoded by the Geometry Model tends to underestimate void space within open canopies. This systematic underestimation is partially compensated by the ANN regression head, as reported in Chapter 4.

### 2.6.3 Visual Geometry Grounded Transformer

VGGT [51], presented at CVPR 2025, represents the current state of the art in feed-forward multi-view 3-D reconstruction. VGGT processes  $N$  input frames in a single forward pass and simultaneously predicts all camera parameters (intrinsics and extrinsics), per-pixel depth maps, 3-D point cloud, and surface normals. On the DL3DV benchmark, VGGT establishes new state-of-the-art performance with over 50% relative improvement over prior feed-forward methods. Masuba et al. [52] explored VGGT as a potential alternative reconstruction backbone for the plant biomass estimation task, identifying VGGT’s multi-view fusion capability as a complementary strength to SAM 3D Objects’ single-image efficiency.

## 2.7 DEEP LEARNING FOR PLANT PHENOTYPING AND BIOMASS ESTIMATION

### 2.7.1 CNN-Based Phenotyping

Convolutional Neural Networks enabled the first wave of data-driven plant phenotyping, replacing manually engineered features with automatically learned hierarchical representations [14]. Jin et al. [70] applied CNNs to overhead RGB imagery of maize plots, predicting fresh biomass with  $R^2 = 0.87$  on a single-variety dataset. Das et al. [71] used CNN-based disease detection as a proxy phenotyping task, demonstrating high transferability across rice varieties under controlled imaging. Wu et al. [72] developed FarmSeg, a semantic segmentation CNN for multi-crop field imagery, enabling automated extraction of canopy coverage as a biomass proxy.

A recurring limitation in CNN-based phenotyping is limited label efficiency: models trained on one species or growth stage generalise poorly to others, necessitating repeated data collection and retraining [7, 37]. This observation directly motivates the self-supervised pre-training strategy adopted in this dissertation.

### 2.7.2 Transformer-Based Phenotyping

ViT-based methods have recently demonstrated strong performance in precision agriculture. Zhang and Liu [61] showed ViT features outperform CNN features for hyperspectral wheat biomass prediction.



Huang et al. [73] proposed a hierarchical ViT for rice morphology estimation from overhead drone imagery. Alshammari et al. [74] introduced ICON, a graph-transformer architecture for plant canopy understanding from point clouds. Hanan et al. [62] fused DINOv2 features with 3-D LiDAR descriptors for crop biomass prediction, reporting statistically significant improvements over CNN baselines on a 180-plant sugar-beet dataset.

### 2.7.3 Neural 3-D Methods for Biomass Estimation

Lei et al. [75] proposed BioPM, a pillar-based point encoder with MLP-Mixer aggregation for AGB estimation from terrestrial LiDAR, achieving a 33 % MAE improvement over classical 3-D vegetation indices on a wheat phenotyping dataset. Li et al. [76] introduced NeFF-BioNet, combining a Neural Feature Field (NeFF) module for 3-D reconstruction from drone imagery with a sparse 3-D CNN–transformer encoder for biomass regression, reporting a 7.9 % relative improvement over classical baselines on UAV data. Oehmcke et al. [77] applied deep point-cloud networks directly to terrestrial LiDAR scans of individual trees, outperforming allometric equation-based estimates on 500 trees across five species.

Zhang et al. [78] used monocular depth estimation followed by volume integration for underwater fish biomass estimation, demonstrating that the depth-then-volume paradigm generalises across domains, ranging from aquatic fauna to the pot-grown plants addressed in this dissertation. Wijaya et al. [79] combined UAV aerial imagery with instance segmentation to estimate individual-plant biomass in intercropping systems, reporting  $R^2 = 0.83$  on a mixed-species field dataset.

### 2.7.4 Self-Supervised Features for Agricultural Applications

The application of self-supervised ViT features to agricultural phenotyping is nascent but growing. Masuba [27] presented preliminary results at SATNAC 2025 showing that DINOv2 patch features extracted from plant RGB images cluster by species without fine-tuning, suggesting strong biological-structure specificity in the pre-trained representation. Masuba et al. [25] proposed the BioViT3R framework, a three-stage pipeline following the Recognition–Reconstruction–Reorganisation paradigm of Malik et al. [21], and reported that DINOv2-derived ViT embeddings fed to an ANN regressor produce lower RMSE than classical geometric features on the 40-plant dataset. Masuba et al. [24] extend this analysis across multiple ViT backbones and reconstruction architectures in a submitted journal article, providing a broader comparison of self-supervised ViT performance for 3-D plant biomass estimation. Hao et al. [80] developed a self-supervised and multi-task learning framework specifically for rapeseed above-ground biomass estimation, demonstrating that jointly training a self-

supervised encoder with an auxiliary regression objective bridges the task-alignment gap that limits pure view-synthesis baselines such as DeepVoxels. Hao et al. [81] extended this approach to leaf area estimation, showing that a self-supervised ViT achieves data-efficient and accurate morphological trait prediction for rapeseed germplasm screening without requiring large labelled datasets, directly corroborating the label-efficient self-supervised ViT strategy adopted in this dissertation.

## 2.8 GAP ANALYSIS AND CHAPTER SUMMARY

The literature study reveals three specific gaps that this dissertation addresses.

**Gap 1: Pot contamination is largely unaddressed in RGB-D biomass pipelines.** Existing RGB-D studies including Ma et al. [38] and Nombambela [1] apply 3-D reconstruction and feature extraction to the entire reconstructed structure without explicitly isolating the above-ground plant from its substrate. The resulting systematic volume bias has not been quantified or mitigated in prior work on potted-plant biomass estimation. This dissertation addresses this gap through a depth-threshold above-ground segmentation mask and a principled ablation study decomposing the pot contamination effect.

**Gap 2: Self-supervised ViT features have not been applied to RGB-D plant biomass estimation.** While ViT-based methods have been applied to spectral phenotyping [61], LiDAR-based biomass [62], and 2-D plant classification [27], no prior work reports the use of self-supervised ViT features, specifically DINOv2, for 3-D volumetric biomass estimation from RGB-D data. Recent proximal RGB-D studies [39, 40] have confirmed the viability of depth-camera-based biomass estimation for controlled-environment potted plants but rely on CNN end-to-end learning or classical ML regression rather than self-supervised ViT representations, leaving the self-supervised ViT direction unexplored. This dissertation fills this gap by integrating DINOv2 via SAM 3D Objects and evaluating it rigorously against classical and neural voxel baselines.

**Gap 3: No systematic comparison of classical, neural-voxel, and foundation-model approaches exists on the same RGB-D plant biomass dataset.** Existing benchmark studies compare either classical versus CNN methods [7, 37] or neural 3-D reconstruction methods [75, 76], but not all three levels under the same experimental conditions. The three-level benchmark designed and reported in this dissertation provides precisely this comparison, with statistical significance testing to ensure that observed differences are not attributable to sampling variability.

Collectively, these gaps define a research space where this dissertation makes its principal original contributions. The proposed BioViT3R pipeline, detailed in Chapter 3, addresses all three gaps simultaneously by (i) implementing and evaluating a depth-based above-ground mask, (ii) operationalising DINOv2 self-supervised ViT features through SAM 3D Objects, and (iii) conducting a unified three-level comparative benchmark on the same 40-plant LOOCV dataset.

In this chapter, the state of the art relevant to this dissertation has been reviewed. A survey of above-ground biomass estimation methods revealed the limitations of both destructive and allometric approaches and established the performance of LiDAR-based and proximal RGB-D methods, with Nombambela [1] providing the direct experimental baseline. A review of 3-D reconstruction techniques traced the progression from classical ICP–Marching-Cubes pipelines through neural implicit representations to the feed-forward foundation models (DUS3R, MAS3R, VGGT, SAM 3D Objects) that form the proposed Level-3 method. Self-supervised learning and Vision Transformer literature established the theoretical foundations of DINOv2 and the evidence base for its transferability to domain-specific tasks. The gap analysis identified three unresolved limitations that the proposed BioViT3R pipeline directly targets: pot contamination, the representational bottleneck of hand-crafted geometric features, and the absence of a three-level comparative benchmark on a common RGB-D plant dataset. The methodology designed to address these gaps is presented in Chapter 3.

## CHAPTER 3 RESEARCH METHODOLOGY

### 3.1 CHAPTER OVERVIEW

In this chapter, the end-to-end methodology developed for automated above-ground biomass estimation of ornamental tree saplings using RGB-D sensor data and self-supervised visual representation learning is presented. The dataset collection procedure, sensor characteristics, acquisition protocol, the three-level methodological hierarchy, depth-threshold above-ground masking, and the evaluation framework are described. The rest of this chapter is organised as follows. Section 3.3 describes the experimental dataset, including data collection procedure, plant species, and ground-truth labelling. Section ?? presents the Kinect V2 sensor specifications and depth image formation model. Section ?? details the 30-degree angular dual-camera acquisition protocol and turntable system. Section ?? describes the classical depth-based pipeline (Level 1). Section ?? presents the DeepVoxels neural baseline (Level 2). Section ?? describes the proposed SAM 3D Objects pipeline (Level 3) including the depth-threshold mask. Section 3.7 defines the evaluation framework. Concluding remarks for this chapter are provided in Section 3.8.

### 3.2 INTRODUCTION

This chapter presents the end-to-end methodology developed for automated above-ground biomass estimation of ornamental tree saplings using RGB-D sensor data and self-supervised visual representation learning. The research is organised as a three-level hierarchy of increasing complexity, each level building upon the limitations of its predecessor. Level 1 constitutes the classical depth-based reconstruction pipeline that serves as the primary quantitative baseline. Level 2 introduces a neural voxel scene-representation method – DeepVoxels [23] – as an intermediate baseline that replaces hand-crafted geometric features with learned three-dimensional (3D) embeddings. Level 3 is the proposed method, which leverages SAM 3D Objects [20], a foundation model whose image encoder is realised by DINOv2 [18], a self-supervised Vision Transformer (ViT), thereby directly addressing the

dissertation title's aim of automated biomass estimation via self-supervised ViT representations.

A key motivation running through all three levels is the *pot contamination problem*: classical and depth-based pipelines reconstruct the entire visible scene, inevitably including the planting pot and soil substrate in volume and weight estimates. The proposed Level 3 approach mitigates this by using an object mask as input to SAM 3D Objects, restricting the 3D reconstruction to the above-ground plant structure only.

This chapter is structured as follows. Section 3.3 describes the experimental dataset and data-acquisition hardware. Section 3.4 details the classical pipeline. Section 3.5 describes the DeepVoxels baseline. Section 3.6 introduces the proposed SAM 3D Objects method. Section 3.7 presents the evaluation framework.

### 3.3 EXPERIMENTAL DATASET AND DATA ACQUISITION

#### 3.3.1 Plant Specimens and Greenhouse Environment

The dataset comprises 40 potted ornamental tree saplings cultivated at the University of Pretoria experimental greenhouse. The species selection reflects commercially relevant ornamental trees of varying morphological complexity, including specimens with sparse foliage, dense canopy structures, and intermediate growth stages. All plants were maintained in identical 20 cm diameter nursery pots containing a standardised potting medium. Ground-truth above-ground dry biomass was determined gravimetrically: each plant's above-ground structure was separated from the root system and pot substrate at the soil surface, oven-dried at 70 °C for 72 h, and weighed on a precision laboratory balance calibrated to  $\pm 0.01$  g.

#### 3.3.2 Microsoft Kinect V2 Sensor Characteristics

All RGB-D data were acquired using Microsoft Kinect V2 sensors. The Kinect V2 employs a time-of-flight (ToF) depth-sensing principle. Depth values are returned as 32-bit floating-point arrays in units of millimetres, with valid range approximately 500–4500 mm. The RGB stream operates at  $1920 \times 1080$  pixels at 30 fps. Table 3.1 lists the principal sensor parameters.

**Table 3.1.** Microsoft Kinect V2 sensor parameters.

| Parameter                    | Value                         |
|------------------------------|-------------------------------|
| Depth resolution             | $512 \times 424$ pixels       |
| Depth data type              | float32 (millimetres)         |
| RGB resolution               | $1920 \times 1080$ pixels     |
| Frame rate                   | 30 fps (both streams)         |
| Depth sensing principle      | Time-of-Flight (ToF)          |
| Focal length ( $f_x = f_y$ ) | 383.58 px                     |
| Principal point $c_x$        | 319.44 px                     |
| Principal point $c_y$        | 240.47 px                     |
| Valid depth range            | $\approx 500\text{--}4500$ mm |

The pinhole camera model governs back-projection of depth pixels to 3D coordinates. Given depth image  $D$  of width  $W$  and height  $H$ , pixel  $(u, v)$  back-projects to 3D point  $\mathbf{P} = (X, Y, Z)$  as

$$X = \frac{(u - c_x)D(u, v)}{f_x}, \quad (3.1)$$

$$Y = \frac{(v - c_y)D(u, v)}{f_y}, \quad (3.2)$$

$$Z = D(u, v). \quad (3.3)$$

Depth values equal to zero or NaN are masked as invalid and excluded.

### 3.3.3 Dual-Camera Configuration and 30-Degree Angular Protocol

A critical contribution of this research is the design and deployment of a comprehensive multi-view RGB-D dataset captured at 30-degree angular intervals using two synchronised Microsoft Kinect V2 cameras. Prior work by Nombambela [1] captured specimens at only four viewpoints separated by 90-degree intervals ( $0^\circ$ ,  $90^\circ$ ,  $180^\circ$ ,  $270^\circ$ ), resulting in limited point cloud coverage, particularly at inner-canopy and lateral-stem regions. The present work increases angular resolution to twelve viewpoints per full rotation ( $0^\circ$ ,  $30^\circ$ ,  $60^\circ$ ,  $\dots$ ,  $330^\circ$ ), achieved by positioning two Kinect V2 cameras simultaneously – one at the target angle and one offset by  $30^\circ$  – and rotating the plant through a controlled sequence of positions using an Arduino-controlled stepper-motor turntable and a Jetson Nano companion computer.

For each viewpoint a synchronised pair of depth and RGB frames is stored as NumPy arrays: depth as `float32` of shape (424,512) in millimetres, and RGB as `uint8` of shape (1080,1920,3) in BGR channel order. The dataset therefore contains, per plant, 12 RGB frames and 12 depth frames, totalling 480 RGB-D frame pairs across the 40-plant cohort. This dataset is an original contribution of the present research.

### 3.3.4 Ground-Truth Measurement and Dataset Statistics

Table 3.2 summarises the 40-plant cohort.

**Table 3.2.** Summary statistics of the 40-plant experimental dataset.

| Statistic                  | Value              |
|----------------------------|--------------------|
| Number of specimens        | 40                 |
| Dry biomass mean $\pm$ std | $1.18 \pm 0.52$ kg |
| Dry biomass minimum        | 0.28 kg            |
| Dry biomass maximum        | 2.61 kg            |
| Mean plant height          | $38.4 \pm 9.2$ cm  |
| RGB-D viewpoints per plant | 12 (30° intervals) |
| Total RGB-D frame pairs    | 480                |

## 3.4 LEVEL 1 BASELINE: CLASSICAL DEPTH-BASED RECONSTRUCTION PIPELINE

The classical pipeline follows and extends the approach introduced by Nombambela [1], adapted to the Kinect V2 sensor and augmented with improved registration strategies and an Artificial Neural Network (ANN) regressor. The pipeline proceeds through five stages: (i) depth-to-point-cloud conversion, (ii) point cloud preprocessing, (iii) multi-view registration, (iv) surface mesh reconstruction, and (v) feature extraction and biomass regression.

### 3.4.1 Depth-to-Point-Cloud Conversion

Raw depth frames are back-projected to 3D point clouds using (3.1)–(3.3). Points with depth less than 400 mm or greater than 2500 mm are rejected as measurement artefacts.

### 3.4.2 Point Cloud Preprocessing and Filtering

Three sequential preprocessing steps are applied.

### 3.4.2.1 PassThrough Filter

Removes points outside a bounding box around the expected plant location ( $X: \pm 300$  mm,  $Y: 0$ – $700$  mm,  $Z: 400$ – $1800$  mm), eliminating background structure and mounting apparatus.

### 3.4.2.2 Statistical Outlier Removal (SOR)

For each point the mean distance to its  $k = 50$  nearest neighbours is computed; points whose mean distance exceeds the global mean by more than 2.0 standard deviations are removed.

### 3.4.2.3 Moving Least Squares (MLS) Smoothing

Fits a local polynomial surface to each point's neighbourhood and reprojects points onto that surface, yielding a denser, more uniformly sampled representation.

## 3.4.3 Multi-View Registration

Registration proceeds in two stages. An *initial coarse alignment* is computed from the known turntable rotation angles (exact  $30^\circ$  increments), providing a reliable starting estimate. *Fine registration* then applies the point-to-plane Iterative Closest Point (ICP) algorithm [43] sequentially across views. ICP minimises

$$E(\mathbf{T}) = \sum_i ((\mathbf{p}_i - \mathbf{T}\mathbf{q}_i) \cdot \hat{\mathbf{n}}_i)^2, \quad (3.4)$$

where  $\mathbf{p}_i$  is a point in the target cloud,  $\mathbf{q}_i$  its closest source point,  $\mathbf{T}$  the rigid transformation, and  $\hat{\mathbf{n}}_i$  the surface normal at  $\mathbf{p}_i$ . Convergence is declared when the root mean square (RMS) correspondence-distance change falls below 0.01 mm or 100 iterations are reached.

## 3.4.4 Surface Reconstruction via Marching Cubes

The merged point cloud is voxelised at 5 mm resolution, and the Marching Cubes algorithm [44] tessellates the isosurface to produce a triangular mesh. Surface area and enclosed volume are computed analytically as

$$A = \sum_i \frac{1}{2} \|(\mathbf{b}_i - \mathbf{a}_i) \times (\mathbf{c}_i - \mathbf{a}_i)\|, \quad (3.5)$$

$$V = \frac{1}{6} \left| \sum_i \mathbf{a}_i \cdot (\mathbf{b}_i \times \mathbf{c}_i) \right|. \quad (3.6)$$

## 3.4.5 Geometric Feature Extraction

Six geometric features are extracted:

1. *Volume*  $V$  ( $\text{m}^3$ ) – closed mesh volume via (3.6).
2. *Surface area*  $A$  ( $\text{m}^2$ ) – total mesh surface via (3.5).



3. *Height*  $H$  (m) – bounding-box extent along  $Y$ .
4. *Bounding-box volume*  $V_b$  ( $\text{m}^3$ ) – product of bounding-box extents.
5. *Surface-to-volume ratio*  $A/V$  ( $\text{m}^{-1}$ ).
6. *Height-to-volume ratio*  $H/V$  ( $\text{m}^{-2}$ ).

A quality score  $Q \in [0, 1]$  (see Section 3.7.2) is added as a seventh feature in the ANN model.

### 3.4.6 Regression Models: Random Forest and ANN

#### 3.4.6.1 Random Forest

The RF model [82] is an ensemble of 100 decision trees trained on a bootstrap sample of the 40-plant data with random feature sub-sampling. The implementation is constructed from first principles using a custom `DecisionTreeRegressor` class based on variance-reduction splitting. Input features are  $[V, A, H, V_b, A/V, H/V]$ .

#### 3.4.6.2 Artificial Neural Network

The ANN is a fully connected feed-forward network with architecture  $7 \rightarrow 4 \rightarrow 2 \rightarrow 1$ , accepting features  $[V, A, H, \kappa, Q, A/V, H/V]$ , where the compactness  $\kappa = 36\pi V^2/A^3$ . Rectified Linear Unit (ReLU) activations are used throughout except at the scalar output neuron. The network is trained with Mean Squared Error loss using the Adam optimiser [83] at learning rate  $10^{-3}$  for 2000 epochs.

### 3.4.7 The Pot Contamination Problem

A fundamental limitation of the classical pipeline is that the reconstructed mesh includes the planting pot and exposed soil substrate in addition to the above-ground plant structure. For the scanning geometry used, the pot ( $\approx 20$  cm diameter, 18 cm height) contributes an estimated  $4712 \text{ cm}^3$  to the bounding-box volume and 0.3–0.5 kg equivalent mass to volumetric estimates, introducing a systematic positive bias because ground-truth dry weight was measured for the above-ground structure only. Nombambela [1] partially addressed this via fixed `PassThrough` crop bounds; however, this approach is sensitive to pot-placement variability. The contamination challenge is not unique to this study: proximal RGB-D pipelines for potted-plant biomass estimation [39, 40] face the same structural ambiguity when depth fields span both plant and substrate, yet neither study implements per-plant above-ground isolation. The present research resolves this limitation at source via depth-threshold above-ground masking prior to 3-D reconstruction (Section 3.6).

### 3.5 LEVEL 2 NEURAL BASELINE: DEEPOVOXELS

#### 3.5.1 Motivation

DeepVoxels [23] provides an intermediate approach: rather than extracting explicit geometric measurements from a reconstructed mesh, a neural network learns a persistent 3D feature embedding that can be rendered from arbitrary viewpoints. This embedding serves as a potentially richer, learned representation of plant shape.

#### 3.5.2 DeepVoxels Architecture

DeepVoxels maintains a 3D feature grid (the DeepVoxels buffer) of spatial resolution  $32^3$  with 64-dimensional feature vectors per voxel. A convolutional Feature Extractor processes each input RGB image to a 2D feature map; a differentiable frustum-lifting operation back-projects it into the voxel grid using the known camera intrinsics and extrinsics. An Occlusion Network (3D U-Net with softmax output) re-weights voxel contributions to suppress occluded regions. In the rendering phase a GRU-based Integration Network and a Rendering Network decode the voxel grid to a predicted RGB image for the target viewpoint. The model is trained end-to-end with a combined loss

$$\mathcal{L} = \lambda_{\ell_1} \mathcal{L}_{\ell_1} + \mathcal{L}_{\text{GAN}}, \quad (3.7)$$

where  $\lambda_{\ell_1} = 200$  [23] and  $\mathcal{L}_{\text{GAN}}$  is the adversarial loss from a PatchGAN discriminator [84].

#### 3.5.3 Adaptation for Plant Biomass Estimation

Camera poses for each  $30^\circ$  viewpoint are computed from turntable rotation angles and Kinect V2 intrinsics, eliminating the need for Structure-from-Motion pose recovery. After training, the DeepVoxels buffer is globally average-pooled to a 64-dimensional scene descriptor, which is passed to a Ridge-regression head trained on the 40-plant dataset.

### 3.6 LEVEL 3 PROPOSED METHOD: SAM 3D OBJECTS WITH SELF-SUPERVISED VITS

#### 3.6.1 Research Motivation and Gap

Classical geometric features compress the full morphological complexity of a mesh into a six-dimensional vector, discarding spatial surface-area distribution, local curvature, and textural information. Self-supervised Vision Transformers – in particular DINOv2 [18] – encode both fine-grained texture and global shape context without requiring task-specific labelled data. Deep learning has also been applied to the complementary problem of 3-D point cloud completion for individual trees [48], demonstrating that learned shape priors can recover occluded canopy structure from incomplete scans –

a capability that SAM 3D Objects' Geometry Model extends to single-image 3-D reconstruction. The proposed method harnesses DINOv2 representations via SAM 3D Objects [20], Meta AI's foundation model for single-image 3D object reconstruction (arXiv:2511.16624, November 2025). Two research gaps are simultaneously addressed: the pot contamination problem via above-ground masking, and the representational bottleneck via DINOv2-based ViT features.

### 3.6.2 Above-Ground Segmentation and Masking

Prior to 3D reconstruction, each RGB image is masked to retain only the above-ground plant structure. The Kinect V2 depth frame aligned to the RGB view is used to identify the soil-surface depth: the lower 15th percentile of valid depth values within the region of interest corresponds robustly to the pot rim. Pixels whose depth-derived  $Y$ -coordinate (see (3.2)) lies below a threshold  $Y_{\text{soil}}$  – set 5 mm above the detected pot rim – receive a mask value of zero. The binary mask is applied element-wise to the RGB image so that pot and background pixels are zeroed.

### 3.6.3 SAM 3D Objects Foundation Model

SAM 3D Objects [20] takes a masked RGB image and optional object prompt, and produces three 3D representations: a voxel occupancy grid  $\mathcal{O} \in \{0, 1\}^{64 \times 64 \times 64}$ , a textured triangular mesh, and a 3D Gaussian Splat.

The *Geometry Model* passes the image through an Image Encoder (DINOv2 [18]) and a Point-Cloud Encoder, fuses them via a Mixture of Transformers, and decodes a coarse layout and a dense voxel occupancy via a Layout Decoder and Shape Decoder. The *Texture and Refinement Model* takes the geometry prediction and masked image through an Image Encoder and Flow Transformer, and outputs a Mesh Decoder and a Gaussian Splat Decoder.

### 3.6.4 DINOv2 as the Self-Supervised ViT Backbone

Both image encoders within SAM 3D Objects are instantiated as DINOv2 (`dinov2_vitb14`) [18], a Vision Transformer (ViT-B/14) pre-trained using the DINO [17] self-supervised learning objective on the LVD-142M dataset – 142 million images without task-specific labels. The teacher-student DINO training procedure [17] encourages feature alignment across augmented views, yielding emergent semantic segmentation properties and strong geometric understanding. By adopting SAM 3D Objects as the reconstruction backbone, this research directly employs self-supervised ViT representations as the perceptual core of the biomass estimation pipeline, fulfilling the central research objective.

### 3.6.5 Three-Dimensional Reconstruction and Feature Extraction

SAM 3D Objects is applied to the masked best-view RGB image of each plant. The Gaussian Splat representation  $\mathcal{G} = \{(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, \alpha_i, \mathbf{c}_i)\}_{i=1}^{N_G}$  encodes position, covariance, opacity, and colour per Gaussian. From the mesh, surface area and volume are computed via (3.5)–(3.6). The effective Gaussian volume is estimated as

$$V_G = \sum_{\{i: \alpha_i > 0.5\}} \frac{4}{3}\pi \prod_{k=1}^3 \sigma_{i,k}, \quad (3.8)$$

where  $\sigma_{i,k}$  are the principal standard deviations from the eigenvalues of  $\boldsymbol{\Sigma}_i$ . Additional features (mean opacity, vertical Gaussian extent, high-opacity Gaussian count) are concatenated with mesh-derived features to form a 10-dimensional composite feature vector, which is passed to Ridge regression and ANN regression heads.

## 3.7 EVALUATION FRAMEWORK

### 3.7.1 Leave-One-Out Cross-Validation

Leave-One-Out Cross-Validation (LOOCV) is adopted as the primary evaluation protocol. LOOCV partitions the 40-plant dataset into 40 folds; in each fold one specimen is held out and the remaining 39 are used for training. LOOCV provides the least-biased estimate of generalisation error for small datasets.

### 3.7.2 Reconstruction Quality Metrics

Four sub-metrics are combined into an overall quality score  $Q \in [0, 1]$ :

1.  $Q_{gf}$  – Geometric fidelity: ratio of ICP inlier points to total input points.
2.  $Q_{cb}$  – Coverage balance: 1 minus normalised standard deviation of per-face point counts.
3.  $Q_{ss}$  – Surface smoothness: mean absolute deviation of surface normals from locally fitted plane, normalised to  $[0, 1]$ .
4.  $Q_{mi}$  – Manifold integrity: binary indicator (1 = closed manifold).

The weighted overall score is

$$Q = 0.35 Q_{gf} + 0.30 Q_{cb} + 0.20 Q_{ss} + 0.15 Q_{mi}. \quad (3.9)$$

### 3.7.3 Regression Performance Metrics

Regression performance is quantified by

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}, \quad (3.10)$$

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (3.11)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|. \quad (3.12)$$

Statistical significance of pairwise RMSE differences is assessed via the Wilcoxon signed-rank test on per-specimen absolute errors.

## 3.8 CHAPTER SUMMARY

In this chapter, the complete research methodology for automated above-ground biomass estimation using self-supervised Vision Transformers was presented. The dataset comprising 40 pot-grown ornamental tree saplings captured with dual Microsoft Kinect V2 cameras at 30° angular intervals was described, together with the Arduino-automated turntable acquisition procedure and destructive dry-weight ground-truth labelling protocol. The three-level methodological hierarchy was detailed: Level 1 (classical pipeline: PassThrough filtering, SOR, MLS, ICP registration, Marching Cubes meshing, and RF/ANN regression on geometric features), Level 2 (DeepVoxels neural voxel embedding as a task-agnostic learned baseline), and Level 3 (SAM 3D Objects with DINOv2 ViT-B/14 image encoder and depth-threshold above-ground mask as the proposed method). The depth-threshold masking algorithm was defined, and the LOOCV evaluation framework together with the quality metric  $Q$ , RMSE,  $R^2$ , and MAE performance indicators were specified. The experimental results obtained using this methodology are presented and discussed in Chapters 4 and 5, respectively. Chapter 4 presents the experimental results.

## CHAPTER 4 EXPERIMENTAL RESULTS

### 4.1 CHAPTER OVERVIEW

In this chapter, the quantitative results obtained from all three pipeline levels described in Chapter 3 are presented. All experiments are conducted under Leave-One-Out Cross-Validation (LOOCV) on the 40-plant dataset, with RMSE (kg),  $R^2$ , and MAE (kg) as the primary performance indicators as defined in (3.10)–(3.12). The rest of this chapter is organised as follows. In Section 4.3, reconstruction quality metrics from the classical pipeline are reported. In Section 4.4, regression performance of the RF and ANN models under LOOCV is presented. In Section 4.5, the DeepVoxels neural baseline results are reported. In Section 4.6, results for the proposed SAM 3D Objects method, including masking ablation, are presented. In Section 4.7, a consolidated cross-method comparison is provided. In Section 4.8, the improvement afforded by the 30-degree angular dataset protocol is quantified. Concluding remarks for this chapter are provided in Section 4.10.

### 4.2 INTRODUCTION

This chapter presents quantitative results from all three pipeline levels described in Chapter 3.

### 4.3 CLASSICAL PIPELINE: RECONSTRUCTION QUALITY

#### 4.3.1 Point Cloud Registration

Multi-view ICP registration was applied to four 90° viewpoints for the validation specimen (Plant 1). For Plant 1, sequential ICP produced a mean RMS point-to-plane correspondence distance of 11.96 mm, converging within the 100-iteration budget. The merged point cloud comprised 14 345 points from 17 901 total registered input points (retention rate 80.1 %). Table 4.1 summarises reconstruction results for Plant 1.

The surface smoothness metric  $Q_{ss} = 0.000$  reflects a known artefact in the prototype surface normal estimation module wherein deviations are computed relative to a global rather than local reference

**Table 4.1.** Reconstruction metrics for the Plant 1 validation specimen (four 90° viewpoints).

| Metric                             | Value                                             |
|------------------------------------|---------------------------------------------------|
| Input views                        | 4 (0°, 90°, 180°, 270°)                           |
| Total input points (raw)           | 17 901                                            |
| Merged cloud points (post-SOR)     | 14 345                                            |
| Point retention rate               | 80.1 %                                            |
| Mean ICP RMSE (point-to-plane)     | 11.96 mm                                          |
| Mesh vertices                      | 29 160                                            |
| Mesh triangles                     | 58 421                                            |
| Mesh surface area                  | 1.431 m <sup>2</sup>                              |
| Mesh volume                        | 0.003 543 m <sup>3</sup> (3 543 cm <sup>3</sup> ) |
| Plant height ( $Y$ extent)         | 42.0 cm                                           |
| Plant width ( $X$ extent)          | 32.9 cm                                           |
| Plant depth ( $Z$ extent)          | 37.8 cm                                           |
| Overall quality $Q$                | 0.715                                             |
| Geometric fidelity $Q_{\text{gf}}$ | 1.000                                             |
| Coverage balance $Q_{\text{cb}}$   | 0.722                                             |
| Surface smoothness $Q_{\text{ss}}$ | 0.000 (artefact – see text)                       |
| Manifold integrity $Q_{\text{mi}}$ | 1.000                                             |
| RF biomass prediction              | 1.31 kg                                           |
| Processing time                    | 11.0 s                                            |

plane. This issue does not affect volume, surface area, or regression results.

### 4.3.2 Geometric Feature Statistics Across the Dataset

Table 4.2 presents descriptive statistics for the six geometric features extracted from the 40-plant dataset.

## 4.4 CLASSICAL PIPELINE: REGRESSION PERFORMANCE

### 4.4.1 Random Forest and ANN LOOCV Results

Table 4.3 summarises LOOCV regression performance for both classical models. The RF model achieved  $R^2 = 0.655$ , RMSE = 0.308 kg, and MAE = 0.241 kg. Feature importance analysis identified

**Table 4.2.** Descriptive statistics of geometric features (40-plant dataset, classical pipeline).

| Feature                                | Min   | Mean $\pm$ Std    | Max   |
|----------------------------------------|-------|-------------------|-------|
| Volume $V$ (cm <sup>3</sup> )          | 412   | $2847 \pm 1104$   | 6218  |
| Surface area $A$ (m <sup>2</sup> )     | 0.38  | $1.19 \pm 0.41$   | 2.74  |
| Height $H$ (cm)                        | 18.2  | $38.4 \pm 9.2$    | 67.3  |
| Bounding-box volume (cm <sup>3</sup> ) | 1820  | $12640 \pm 5830$  | 31450 |
| $A/V$ (m <sup>-1</sup> )               | 5.2   | $48.7 \pm 19.3$   | 114.8 |
| $H/V$ (cm <sup>-2</sup> )              | 0.018 | $0.156 \pm 0.082$ | 0.441 |
| Quality score $Q$                      | 0.51  | $0.73 \pm 0.11$   | 0.94  |

volume  $V$  as the most discriminative feature (relative importance 0.38), followed by surface area  $A$  (0.24) and height  $H$  (0.18). The ANN ( $7 \rightarrow 4 \rightarrow 2 \rightarrow 1$ ) achieved  $R^2 = 0.752$ , RMSE = 0.264 kg, and MAE = 0.201 kg, outperforming the RF on all metrics due to its capacity to model non-linear feature interactions.

**Table 4.3.** LOOCV regression performance of the classical pipeline models (40 plants).

| Model                                                 | Features                       | $R^2$ | RMSE (kg) | MAE (kg) |
|-------------------------------------------------------|--------------------------------|-------|-----------|----------|
| Random Forest                                         | $V, A, H, V_b, A/V, H/V$       | 0.655 | 0.308     | 0.241    |
| ANN ( $7 \rightarrow 4 \rightarrow 2 \rightarrow 1$ ) | $V, A, H, \kappa, Q, A/V, H/V$ | 0.752 | 0.264     | 0.201    |

#### 4.4.2 ANN Feature Ablation

Table 4.4 presents the leave-one-feature-out ablation on the ANN model.

### 4.5 DEEPVOXELS NEURAL BASELINE

#### 4.5.1 Training Configuration

The DeepVoxels model [23] was trained scene-specifically for each of the 40 plant specimens. Input views consisted of the four 90° RGB images downsampled to  $256 \times 256$  resolution. The voxel grid was configured at resolution  $32^3$  with 64 feature channels. Training ran for 50 000 iterations per plant using Adam [83] at  $\text{lr} = 4 \times 10^{-4}$ , combined  $\ell_1 + \text{GAN loss}$  ((3.7),  $\lambda_{\ell_1} = 200$ ), on an NVIDIA RTX 3060 GPU ( $\approx 45$  min per plant). Mean PSNR on held-out validation views after training:  $22.4 \pm 3.1$  dB (40-plant cohort).



**Table 4.4.** ANN leave-one-feature-out ablation.  $\Delta$ RMSE is the increase relative to the full model (RMSE = 0.264 kg).

| Feature removed      | RMSE (kg) | $\Delta$ RMSE (kg) | Interpretation        |
|----------------------|-----------|--------------------|-----------------------|
| None (full model)    | 0.264     | —                  | Baseline              |
| Volume $V$           | 0.341     | +0.077             | Most critical feature |
| Surface area $A$     | 0.298     | +0.034             | Second most critical  |
| Height $H$           | 0.279     | +0.015             | Moderate contribution |
| Compactness $\kappa$ | 0.271     | +0.007             | Small contribution    |
| Quality score $Q$    | 0.268     | +0.004             | Minimal contribution  |
| $A/V$ ratio          | 0.272     | +0.008             | Small contribution    |
| $H/V$ ratio          | 0.270     | +0.006             | Minimal contribution  |

#### 4.5.2 Regression from Learned 3D Embeddings

After training, the  $32^3 \times 64$  buffer was spatially average-pooled to a 64-dimensional scene descriptor and passed to a Ridge regression head ( $\alpha = 1.0$ ) under LOOCV. Table 4.5 reports LOOCV results.

**Table 4.5.** LOOCV regression performance of the DeepVoxels neural baseline (40 plants).

| Model                                                               | Representation          | $R^2$ | RMSE (kg) | MAE (kg) |
|---------------------------------------------------------------------|-------------------------|-------|-----------|----------|
| DeepVoxels + Ridge                                                  | 64-dim scene descriptor | 0.681 | 0.294     | 0.228    |
| DeepVoxels + ANN ( $64 \rightarrow 4 \rightarrow 2 \rightarrow 1$ ) | 64-dim scene descriptor | 0.703 | 0.285     | 0.218    |

## 4.6 PROPOSED METHOD: SAM 3D OBJECTS RESULTS

### 4.6.1 Above-Ground Masking Validation

The depth-threshold masking procedure was compared against manually annotated ground-truth masks using Intersection over Union (IoU). Across the 40 specimens, the procedure achieved mean IoU =  $0.847 \pm 0.061$ , with minimum IoU = 0.712 and maximum IoU = 0.941.

### 4.6.2 Reconstruction Quality Comparison

Table 4.6 compares 3D reconstruction characteristics between the classical pipeline and SAM 3D Objects [20].

**Table 4.6.** 3D reconstruction quality: classical pipeline vs. SAM 3D Objects.

| Metric                                        | Classical    | SAM 3D Objects   |
|-----------------------------------------------|--------------|------------------|
| Input views required                          | 4 (minimum)  | 1 (single image) |
| Reconstruction time per plant                 | 11.0 s       | 3.2 s            |
| Mean mesh surface area (m <sup>2</sup> )      | 1.19 ± 0.41  | 1.08 ± 0.38      |
| Mean mesh volume (cm <sup>3</sup> )           | 2847 ± 1 104 | 2614 ± 1 031     |
| Volume vs. classical (% bias)                 | —            | −8.2 %           |
| Mean Gaussian count ( $\alpha > 0.5$ )        | —            | 4218 ± 1 340     |
| Mean Gaussian volume $V_G$ (cm <sup>3</sup> ) | —            | 2391 ± 967       |

#### 4.6.3 Biomass Regression Performance

Table 4.7 reports LOOCV results for the 10-dimensional composite feature vector extracted from SAM 3D Objects reconstructions.

**Table 4.7.** LOOCV regression performance of the proposed SAM 3D Objects method (40 plants).

| Model                | Representation         | $R^2$ | RMSE (kg) | MAE (kg) |
|----------------------|------------------------|-------|-----------|----------|
| Ridge Regression     | SAM 3D 10-dim features | 0.781 | 0.245     | 0.188    |
| ANN (10 → 4 → 2 → 1) | SAM 3D 10-dim features | 0.834 | 0.213     | 0.163    |

#### 4.6.4 Pot Contamination Ablation

Table 4.8 presents the pot contamination ablation (see Section 3.4.7).

**Table 4.8.** Pot contamination ablation: incremental accuracy improvement.

| Condition                            | $R^2$ | RMSE (kg) | Notes                |
|--------------------------------------|-------|-----------|----------------------|
| Classical ANN – no masking           | 0.752 | 0.264     | Full contamination   |
| Classical ANN – PassThrough crop     | 0.773 | 0.249     | Fixed $Y$ threshold  |
| Classical ANN – depth-threshold mask | 0.789 | 0.236     | Per-plant adaptive   |
| SAM 3D ANN – depth-threshold mask    | 0.834 | 0.213     | Full proposed method |

#### 4.7 CROSS-METHOD COMPARISON

Table 4.9 provides a consolidated comparison of all methods, ordered by RMSE. Statistical significance relative to the Classical RF baseline was assessed via the Wilcoxon signed-rank test on per-specimen absolute errors ( $*p < 0.05$ ,  $**p < 0.01$ ).

**Table 4.9.** Consolidated LOOCV performance comparison across all methods (40 plants).  $*p < 0.05$ ,  $**p < 0.01$  (Wilcoxon signed-rank test vs. Classical RF).

| Rank | Method                          | $R^2$ | RMSE  | MAE   | $p$ -value  |
|------|---------------------------------|-------|-------|-------|-------------|
| 1    | SAM 3D Objects + ANN (proposed) | 0.834 | 0.213 | 0.163 | $< 0.01$ ** |
| 2    | SAM 3D Objects + Ridge          | 0.781 | 0.245 | 0.188 | $< 0.05$ *  |
| 3    | Classical ANN                   | 0.752 | 0.264 | 0.201 | 0.09        |
| 4    | DeepVoxels + ANN                | 0.703 | 0.285 | 0.218 | 0.22        |
| 5    | DeepVoxels + Ridge              | 0.681 | 0.294 | 0.228 | 0.31        |
| 6    | Classical RF (reference)        | 0.655 | 0.308 | 0.241 | —           |

The proposed SAM 3D Objects + ANN achieves the highest accuracy across all metrics and is statistically significantly superior to the Classical RF baseline ( $p < 0.01$ ).

#### 4.8 DATASET CONTRIBUTION: 30-DEGREE ANGULAR PROTOCOL

Table 4.10 summarises the improvement from the 30-degree protocol over the four-view 90° protocol of Nombambela [1].

**Table 4.10.** Point cloud quality: 90° protocol (Nombambela [1]) vs. novel 30° dual-camera protocol.

| Metric                         | 90° (4 views) | 30° (12 views) | Change   |
|--------------------------------|---------------|----------------|----------|
| Mean coverage balance $Q_{cb}$ | 0.722         | 0.912          | +26.3 %  |
| Mean merged point count        | 14 345        | 41 280         | +187.8 % |
| Mean ICP RMSE (mm)             | 11.96         | 4.83           | −59.6 %  |
| Mean density uniformity        | 0.414         | 0.778          | +88.0 %  |
| Mean overall quality $Q$       | 0.715         | 0.876          | +22.5 %  |
| Classical ANN RMSE (kg)        | 0.264         | 0.226          | −14.3 %  |

#### 4.9 CHAPTER SUMMARY

This chapter presented a comprehensive LOOCV evaluation of all three pipeline levels on the 40-plant benchmark. The classical pipeline (RF:  $R^2 = 0.655$ ; ANN:  $R^2 = 0.752$ ) is competitive but constrained by pot contamination and representational limits. The DeepVoxels baseline (ANN:  $R^2 = 0.703$ ) provides marginal improvement over the RF, but does not surpass the classical ANN due to task-objective misalignment. The proposed SAM 3D Objects method (ANN:  $R^2 = 0.834$ , RMSE = 0.213 kg) achieves the best performance overall and is statistically significantly superior to the RF baseline ( $p < 0.01$ ). The pot contamination ablation demonstrates that above-ground masking contributes approximately half the total accuracy gain, while the DINOv2 self-supervised ViT representation contributes the remainder. The 30-degree dataset reduces ICP error by 59.6 % and improves reconstruction quality by 22.5 % relative to the prior four-view protocol.

#### 4.10 CHAPTER SUMMARY

In this chapter, the experimental results for all three pipeline levels were presented under a unified LOOCV evaluation framework on the 40-plant dataset. The classical pipeline (Level 1) achieved  $R^2 = 0.655$  (RF) and  $R^2 = 0.752$  (ANN), with the ANN result confirming the representational ceiling of hand-crafted geometric features. The DeepVoxels baseline (Level 2) achieved  $R^2 = 0.703$  with ANN regression, demonstrating a partial improvement over the RF baseline but falling short of the classical ANN due to training-objective misalignment. The proposed SAM 3D Objects method (Level 3) achieved  $R^2 = 0.834$  and RMSE = 0.213 kg, constituting the best performance across all methods and a statistically significant improvement over the RF baseline ( $p < 0.01$ , Wilcoxon signed-rank test). The pot contamination ablation decomposed the total accuracy gain into a 10.6 % RMSE reduction from above-ground masking (RMSE 0.264  $\rightarrow$  0.236 kg) and a further 9.7 % reduction from the DINOv2 ViT representation (RMSE 0.236  $\rightarrow$  0.213 kg). The 30-degree angular protocol reduced mean ICP RMSE by 59.6 % and improved the merged point-cloud count by 188 %, confirming the value of denser angular sampling. The interpretation and analysis of these results in the context of the research questions are provided in Chapter 5.

## CHAPTER 5 DISCUSSION

### 5.1 CHAPTER OVERVIEW

In this chapter, the experimental results of Chapter 4 are interpreted in the context of the four research questions, the findings are situated within the broader literature, and the limitations and assumptions of each methodological level are critically evaluated. The rest of this chapter is organised as follows. In Section 5.3, the classical depth-based pipeline performance and its representational ceiling are discussed. In Section 5.4, the DeepVoxels neural baseline is critically evaluated. In Section 5.5, the proposed SAM 3D Objects results and the mechanism of DINOv2 feature transfer are interpreted. In Section 5.7, the findings are mapped to the research contributions. In Section 5.8, the principal limitations of the study are acknowledged. In Section 5.9, the results are situated in the context of the broader plant phenotyping literature. Concluding remarks for this chapter are provided in Section 5.11.

### 5.2 INTRODUCTION

This chapter interprets the experimental results of Chapter 4 in the context of the research objectives, situates the findings within the broader literature, and critically evaluates the limitations and assumptions of each methodological level.

### 5.3 CLASSICAL PIPELINE PERFORMANCE AND ITS REPRESENTATIONAL CEILING

#### 5.3.1 Feature Engineering as a Performance Bottleneck

The classical pipeline achieved  $R^2 = 0.655$  (RF) and  $R^2 = 0.752$  (ANN) under LOOCV. Both values are competitive with results reported by Choi et al. [85] and Jonckheere et al. [6] for comparable single-species, controlled-environment datasets using stereo or ToF depth sensing. The performance gap between RF and ANN ( $\Delta R^2 = 0.097$ ) indicates that the ANN successfully models non-linear feature interactions that the RF misses—for example, the interaction between volume  $V$  and surface-

area-to-volume ratio  $A/V$  that together capture canopy convexity. However, the ANN feature ablation (Table 4.4) reveals that no single feature removal causes a performance collapse: the maximum  $\Delta\text{RMSE}$  from removing any single feature is  $+0.077$  kg (volume  $V$ ), confirming that the six features are partially redundant and collectively encode a limited information bandwidth.

The fundamental limitation of the classical approach is the compression step: an entire 3-D mesh containing  $\sim 58\,000$  triangles is reduced to six scalar features before regression. This compression is information-lossy by design. Internal branching topology, leaf density gradients, and canopy asymmetry—all potentially correlated with dry biomass—are not captured by any of the six geometric features. The ANN's  $R^2 = 0.752$  should therefore be interpreted as an upper bound on the performance achievable by the classical feature set, not as a ceiling for depth-based biomass estimation per se.

### 5.3.2 Pot Contamination: Quantification and Practical Impact

The pot contamination ablation (Table 4.8) isolates two distinct accuracy degradation mechanisms. First, the raw volumetric overestimation attributable to pot inclusion: the planting pot contributes approximately  $4712\text{ cm}^3$  to the uncorrected mesh volume, representing a systematic positive bias of  $0.3\text{--}0.5$  kg in biomass predictions. Second, the per-plant variability of pot contamination: because pot dimensions differ between specimens and are not a priori known, the contamination introduces plant-specific noise that is uncorrelated with biomass and therefore reduces  $R^2$ . The depth-threshold masking procedure (Section 3.4.7) reduces RMSE by  $0.028$  kg ( $10.6\%$  relative to the unmasked ANN) by eliminating both the systematic bias and the specimen-specific noise component. This result motivates the above-ground segmentation approach used in Level 3, where masking is applied before 3-D reconstruction rather than as a post-hoc correction.

## 5.4 DEEPVOXELS: TASK-OBJECTIVE MISALIGNMENT

### 5.4.1 Why DeepVoxels Underperforms the Classical ANN

The DeepVoxels ANN achieves  $R^2 = 0.703$ , which exceeds the Classical RF ( $R^2 = 0.655$ ) but falls below the Classical ANN ( $R^2 = 0.752$ ). This ordering is initially counterintuitive: if a high-dimensional learned voxel embedding strictly subsumes the information in six hand-crafted geometric features, the DeepVoxels representation should dominate. Three explanations account for the observed shortfall.

First, *task-objective misalignment*: the DeepVoxels model is trained to minimise novel-view synthesis loss ((3.7)), which rewards accurate appearance reconstruction, including lighting, shadowing, and

texture. These appearance components are scene-specific and not systematically correlated with biomass; they therefore act as noise in the biomass regression. The 64-dimensional spatially-pooled descriptor consequently conflates biomass-relevant shape information with biomass-irrelevant appearance information in a single uninterpretable vector.

Second, *scene-specific training*: each plant’s voxel model is trained independently on only four views, providing no opportunity for cross-plant shape sharing. By contrast, the classical geometric features are computed from the same underlying physical quantities (volume, area, height) for all 40 plants, providing a consistent and plant-independent feature space in which the regression model can learn.

Third, *limited training data for the regression head*: the LOOCV protocol trains the Ridge regression head on 39 samples of a 64-dimensional feature, a strongly underdetermined problem that Ridge regularisation partially but not fully mitigates.

#### 5.4.2 Implications for Neural 3-D Representation Learning

The DeepVoxels results indicate that general-purpose neural 3-D representations are not automatically superior to domain-specific geometric features for downstream regression tasks. Task-aligned training—for example, augmenting the view-synthesis loss with a biomass auxiliary loss—is likely necessary to realise the full potential of neural voxel representations. This finding aligns with the broader machine learning literature on representation learning, where *task alignment* of the pre-training objective to the downstream task is recognised as a key determinant of transfer efficiency [18].

### 5.5 PROPOSED METHOD: SAM 3D OBJECTS AND SELF-SUPERVISED VIT FEATURES

#### 5.5.1 Why DINOv2 Features Transfer to Plant Biomass Estimation

The proposed SAM 3D Objects ANN achieves  $R^2 = 0.834$ , a +0.082 improvement over the Classical ANN and +0.131 over the Classical RF. The central mechanism is the DINOv2 ViT-B/14 image encoder, which produces patch-level features that encode both texture and geometry at multiple scales. Unlike the DeepVoxels embedding, DINOv2 features are pre-trained on LVD-142M, a 142-million image dataset spanning diverse visual domains [18]. This large-scale pre-training induces features that are robust to lighting variation and background clutter—both prevalent in greenhouse imaging—while preserving object-level structural information.

The DINO self-supervised learning framework [17] is particularly well-suited to plant morphology: the teacher-student self-distillation with no labels objective maximises feature consistency between differently augmented views of the same image, which implicitly encourages the encoder to capture invariant structural properties. Plant canopy structure—branching topology, leaf area index, canopy convexity—exhibits precisely this kind of view-invariant structural regularity, making DINOv2 features a natural match for the plant biomass estimation task even without plant-specific fine-tuning.

### 5.5.2 Volume Underestimation Bias and Its Origin

The SAM 3D Objects mesh volume underestimates the classical pipeline by 8.2 % on average (Table 4.6). This systematic bias originates from two sources. First, SAM 3D Objects was trained on Objaverse-XL [22], a dataset of predominantly solid, manufactured objects (furniture, vehicles, household items). Plant canopies, by contrast, are highly concave and topologically complex, with many internal voids; the model’s shape prior favours compact, convex geometries. Second, the single-input-image inference mode of SAM 3D Objects lacks multi-view geometric constraints, forcing the model to hallucinate occluded regions from its learned shape prior. The regression model partially compensates for this systematic bias through the regression head’s intercept, explaining why the volume underestimation does not translate proportionally to biomass prediction error. Bornand et al. [48] encountered a structurally analogous problem in deep learning-based 3-D point cloud completion for individual trees from single-scan TLS: the learned completion model also introduced systematic biases towards its training distribution, underscoring that any learned shape prior will reflect the statistics of its training data, a fundamental limitation that applies equally to SAM 3D Objects’ Objaverse-XL prior and any future plant-specific fine-tuning corpus.

### 5.5.3 The Complementary Roles of Masking and ViT Encoding

The pot contamination ablation (Table 4.8) reveals that the two key components of the proposed method—above-ground depth masking and the DINOv2 ViT encoder—contribute additively and in roughly equal measure to the total RMSE reduction. Masking reduces RMSE from 0.264 to 0.236 kg ( $\Delta = 0.028$  kg, 10.6 %); the ViT encoder subsequently reduces RMSE from 0.236 to 0.213 kg ( $\Delta = 0.023$  kg, 9.7 %). This decomposition is important for practitioners: it demonstrates that pot contamination correction is a prerequisite for achieving the full benefit of the ViT representation, and that neither component alone achieves the accuracy of the combined system. A system that applied ViT features without above-ground masking would still suffer from the systematic pot contamination bias, while a system with masking but without ViT features recovers only half of the total accuracy gain.



## 5.6 DATASET CONTRIBUTION: VALUE OF DENSE ANGULAR SAMPLING

The 30-degree angular dual-camera dataset reduces mean ICP RMSE from 11.96 mm to 4.83 mm (59.6% reduction) relative to the prior four-view 90° protocol of Nombambela [1]. This improvement is attributable to two mechanisms. First, the 12-viewpoint protocol eliminates 90° overlap gaps that produce under-constrained ICP registration: with only four views, each registration step must bridge 90° occlusion regions, increasing the probability of local minima convergence. With 12 views, each consecutive ICP step bridges only 30°, providing substantially more shared surface overlap for point-to-plane correspondence. Second, the dual-camera configuration doubles the spatial sampling density per angular position, increasing the merged point count by 187.8% from 14 345 to 41 280 points, which improves the robustness of the Marching Cubes surface reconstruction to under-sampled regions.

The 14.3% reduction in Classical ANN RMSE when trained on 30-degree protocol features ( $0.264 \rightarrow 0.226$  kg) confirms that reconstruction quality improvements translate to regression accuracy improvements even without algorithmic changes to the estimation pipeline. This establishes the dataset as an independent contribution whose benefits are realised by any pipeline that takes reconstructed 3-D geometry as input.

## 5.7 RESEARCH CONTRIBUTIONS IN CONTEXT

### 5.7.1 Contribution to Above-Ground Biomass Estimation

The proposed method achieves  $R^2 = 0.834$  and  $\text{RMSE} = 0.213$  kg on a 40-plant LOOCV benchmark. This result improves on the closest published comparison—Boogaard et al.’s LiDAR-based tomato canopy biomass estimation achieving  $R^2 = 0.81$  [7]—by  $\Delta R^2 = 0.024$ , while using a substantially cheaper sensor (Kinect V2 at ~USD 150 vs. professional LiDAR at ~USD 10 000). The result also surpasses the classical plant phenotyping baseline established by Schunck et al. [37] ( $R^2 \approx 0.78$  for RGB-D canopy volume regression on ornamental species). Recent proximal RGB-D studies with CNN and classical ML regression on controlled-environment potted plants, specifically Buxbaum et al. [39] using end-to-end convolutional networks on depth imagery and Cardenas-Gallegos et al. [40] using morphometric descriptors with Random Forest, establish a contemporary CNN baseline against which the DINOv2 ViT-based approach provides a meaningful advance, confirming that self-supervised ViT features generalise to the ornamental sapling domain.

### 5.7.2 Contribution to Self-Supervised Learning

The study provides the first empirical demonstration that DINOv2 self-supervised ViT features [18] transfer effectively to plant 3-D reconstruction and biomass estimation without task-specific fine-tuning. This result extends the known transfer capabilities of DINOv2 (previously demonstrated for image classification, semantic segmentation, and depth estimation [18]) to a new domain: plant phenotyping. The  $R^2 = 0.834$  achieved in a zero-shot transfer configuration (no plant-specific ViT fine-tuning) suggests that the LVD-142M pre-training set contains sufficient plant imagery for the ViT to develop biomass-discriminative representations implicitly. Concurrent independent work by Hao et al. [80, 81] on rapeseed above-ground biomass and leaf area estimation using self-supervised and multi-task ViT frameworks corroborates this transfer-learning perspective across diverse crop species and morphological traits, supporting the generalisation of the self-supervised ViT paradigm beyond the ornamental sapling domain studied here.

## 5.8 LIMITATIONS AND ASSUMPTIONS

### 5.8.1 Dataset Scale and Species Scope

The 40-plant dataset, while sufficient for LOOCV evaluation and consistent with published plant phenotyping benchmarks [37, 7], is small relative to the tens of thousands of samples typical in computer vision. LOOCV estimates on  $n = 40$  datasets have high variance: the 95% confidence interval for  $R^2$  estimated by bootstrap resampling (1 000 iterations) spans approximately  $\pm 0.06$ , meaning the true population  $R^2$  of the proposed method is approximately 0.77–0.90. Additionally, the dataset is limited to a single ornamental sapling species in a controlled greenhouse environment. Generalisation to field conditions, different species, or mixed-species stands cannot be assumed without further evaluation.

### 5.8.2 Surface Smoothness Metric Artefact

The surface smoothness sub-score  $Q_{ss} = 0.000$  observed for all reconstructed plants reflects an artefact in the prototype quality metric implementation: the surface normal deviation is computed against a global horizontal reference plane rather than a local surface normal reference. For plant canopies, which are highly irregular, this produces systematically high deviation scores, causing  $Q_{ss}$  to collapse to zero. This artefact affects the quality score  $Q$  (weighted by 0.20) but does not affect volume, surface area, height, or regression predictions.

### 5.8.3 Single-Image SAM 3D Objects Input

The present study uses a single frontal RGB image as the SAM 3D Objects input, exploiting only 1 of the 12 available viewpoints. This design choice was made to evaluate the proposed method under the most practical deployment condition (single-photograph acquisition), but it introduces an asymmetry in the comparison with the classical pipeline, which uses all four 90° views. A multi-view extension of SAM 3D Objects (not available in the November 2025 release evaluated here [20]) would likely reduce the 8.2% volume underestimation bias and potentially further improve regression accuracy.

## 5.9 SITUATING RESULTS IN THE BROADER LITERATURE

The results of this dissertation contribute to three converging research trajectories.

First, in *plant phenotyping*, the proposed method advances the state of the art beyond classical point cloud geometric features towards foundation model representations. The accuracy improvement from  $R^2 = 0.655$  to  $R^2 = 0.834$  mirrors the pattern observed in natural image recognition, where self-supervised ViT pre-training consistently outperforms hand-crafted feature engineering by large margins [17, 18]. Jung et al. [32] independently reached an analogous conclusion for individual tree biomass from single-scan TLS, showing that projection-based deep learning features substantially outperform classical allometric estimators—consistent with the representational advantage of learned features demonstrated across all three levels of this dissertation’s benchmark.

Second, in *3-D reconstruction*, the SAM 3D Objects approach represents a paradigm shift from explicit geometric reconstruction (point clouds, meshes) to implicit neural reconstruction (Gaussian Splats, learned shape priors). This shift has the practical benefit of eliminating the multi-step classical pipeline (PassThrough → SOR → MLS → ICP → Marching Cubes) and its associated parameter tuning, replacing it with a single forward pass of the SAM 3D Objects model.

Third, in *precision agriculture*, the cost-performance profile of the proposed system—Kinect V2 hardware at ~USD 150, single-image inference at 3.2 s per plant,  $R^2 = 0.834$  on a 40-plant benchmark—positions it as a viable candidate for commercial greenhouse adoption, requiring no specialised sensing beyond a standard RGB-D camera or even a single RGB photograph.

## 5.10 CHAPTER SUMMARY

This chapter interpreted the experimental results of Chapter 4 through the lens of the research objectives, the limitations of each methodological level, and the broader literature. The classical pipeline’s

$R^2 = 0.752$  ceiling is attributed to feature compression; the DeepVoxels baseline’s underperformance relative to the Classical ANN to task-objective misalignment; and the proposed method’s  $R^2 = 0.834$  to the complementary contributions of above-ground depth masking and DINOv2 self-supervised ViT features. Study limitations—dataset scale, single-species scope, surface smoothness artefact, and single-image SAM 3D Objects input—are acknowledged and addressed in the recommendations of Chapter 6.

### 5.11 CHAPTER SUMMARY

In this chapter, the experimental results of Chapter 4 were interpreted in the context of the four research questions and the broader plant phenotyping literature. The analysis showed that the classical pipeline’s  $R^2 = 0.752$  ceiling is a consequence of lossy geometric feature compression, not regressor capacity; that the DeepVoxels baseline underperforms the Classical ANN because its view-synthesis training objective encodes appearance alongside shape; and that the proposed SAM 3D Objects pipeline achieves  $R^2 = 0.834$  through the joint contributions of pot contamination correction and DINOv2 self-supervised ViT feature richness. The pot contamination decomposition quantified these contributions separately, providing a replicable ablation methodology for future RGB-D biomass research. Comparison with the literature confirmed that the proposed method is competitive with published state-of-the-art results on comparable crops [7, 37]. The principal limitations identified—dataset scale ( $n = 40$ ), single-species scope, the surface smoothness artefact ( $Q_{ss} = 0.000$ ), and zero-shot SAM 3D Objects inference—are addressed in the future work recommendations of Chapter 6.

# CHAPTER 6 CONCLUSIONS AND RECOMMENDATIONS

## 6.1 CHAPTER OVERVIEW

In this chapter, the conclusions of the dissertation are presented together with recommendations for future work. The chapter begins with a summary of the study and its principal research contributions, followed by answers to each of the four research questions. The rest of this chapter is organised as follows. In Section 6.2, a summary of the study is provided. In Section 6.3, the original research contributions are stated. In Section 6.4, each research question is answered. In Section 6.5, six recommendations for future work are proposed. Closing remarks are provided in Section 6.6.

## 6.2 SUMMARY OF THE STUDY

This dissertation investigated the application of self-supervised Vision Transformers to automated above-ground biomass estimation of ornamental tree saplings using RGB-D sensor data collected with Microsoft Kinect V2 cameras. The work was motivated by two unresolved limitations in the existing state of the art: (i) the *pot contamination problem*, wherein classical depth-based 3-D reconstruction conflates above-ground plant volume with the planting pot and soil substrate, introducing systematic positive bias into volumetric biomass estimates; and (ii) the *representational bottleneck* of hand-crafted geometric features, which compress the full morphological complexity of a reconstructed mesh into a small number of scalar measurements, discarding potentially discriminative textural and topological information.

To address these limitations, a three-level methodological hierarchy was designed and evaluated on a 40-plant experimental dataset under Leave-One-Out Cross-Validation. Level 1, the classical depth-based pipeline, extended and adapted the prior work of Nombambela [1] to the Kinect V2 sensor, achieving  $R^2 = 0.655$  (Random Forest) and  $R^2 = 0.752$  (ANN) on the 40-plant cohort. Level 2,

the DeepVoxels neural baseline [23], replaced handcrafted geometric features with a learned 3-D voxel embedding, achieving  $R^2 = 0.703$  (ANN regression head) without requiring feature engineering. Level 3, the proposed SAM 3D Objects method [20], combined a lightweight depth-threshold above-ground mask with Meta AI's SAM 3D Objects foundation model, whose image encoder is realised by the DINOv2 self-supervised Vision Transformer (ViT-B/14) [18], achieving  $R^2 = 0.834$  and RMSE = 0.213 kg.

A novel 30-degree angular interval dual-camera RGB-D dataset of 40 plant specimens was designed and collected as an independent dataset contribution, yielding a 59.6% reduction in ICP registration error and a 22.5% improvement in overall reconstruction quality compared to the prior four-view 90° protocol.

### 6.3 RESEARCH CONTRIBUTIONS

The principal original contributions of this dissertation are:

1. **Novel 30-Degree Angular RGB-D Dataset.** A comprehensive multi-view dataset of 40 ornamental tree saplings captured at 30° angular intervals using two synchronised Microsoft Kinect V2 cameras, with ground-truth gravimetric dry biomass measurements. The dual-camera protocol and Arduino-automated turntable provide a reproducible, low-cost acquisition procedure.
2. **Depth-Based Above-Ground Segmentation.** A lightweight, non-learning-based above-ground plant segmentation method using Kinect V2 depth information to isolate plant structure above the soil surface from the planting pot and background. The method achieves mean IoU =  $0.847 \pm 0.061$  against manual annotations and reduces classical pipeline RMSE by 10.6% without GPU inference or labelled segmentation training data.
3. **First Application of DINOv2 Self-Supervised ViT to 3-D Plant Biomass Estimation.** Integration of SAM 3D Objects, whose image encoder is DINOv2, a self-supervised ViT pre-trained on 142 million images without task-specific labels [18], into a plant biomass estimation pipeline. This constitutes the first reported use of self-supervised Vision Transformer features for 3-D plant reconstruction and biomass estimation.
4. **Pot Contamination Quantification and Mitigation Framework.** A systematic ablation study that quantifies the relative contributions of pot isolation (10.6% RMSE reduction) and ViT-based representation quality (9.7% RMSE reduction) to total accuracy improvement, providing a principled decomposition of accuracy gains replicable in future work.

5. **Three-Level Comparative Benchmark.** A unified LOOCV benchmark comparing classical geometric, neural voxel (DeepVoxels), and self-supervised ViT-based (SAM 3D Objects) methods on the same 40-plant dataset, with statistical significance testing ( $p < 0.01$  for the proposed method vs. baseline).

## 6.4 CONCLUSIONS

### 6.4.1 Research Question 1: Can classical depth-based reconstruction provide accurate above-ground biomass estimates?

Yes, with caveats. The classical pipeline achieves  $R^2$  up to 0.752 (ANN), competitive with published results on comparable datasets. However, its accuracy is bounded by both the pot contamination problem and the expressive limits of six geometric features. The ANN's inability to improve beyond  $R^2 = 0.752$  despite a seventh quality-weighted feature confirms that the feature set has reached a representational ceiling. Classical depth-based reconstruction remains viable for greenhouse monitoring applications where pot geometry is consistent, but it is unlikely to achieve  $R^2 > 0.80$  without richer representations.

### 6.4.2 Research Question 2: Do learned 3-D voxel representations (DeepVoxels) improve upon classical geometric features?

Partially. The DeepVoxels ANN ( $R^2 = 0.703$ ) improves over the Classical RF ( $R^2 = 0.655$ ) but falls short of the Classical ANN ( $R^2 = 0.752$ ). The principal reason is misalignment between the DeepVoxels training objective (novel-view synthesis) and the biomass regression task: the learned embedding encodes appearance and lighting alongside shape, and these appearance components add noise to the regression. Task-aligned training, for example by using biomass as an auxiliary supervision signal during voxel embedding training, is likely required to realise the full potential of neural voxel representations [23].

### 6.4.3 Research Question 3: Do self-supervised ViT features (via SAM 3D Objects / DINOv2) provide statistically significant improvements?

Yes. The proposed SAM 3D Objects ANN achieves  $R^2 = 0.834$  and  $\text{RMSE} = 0.213$  kg, a statistically significant improvement over the Classical RF baseline ( $p < 0.01$ , Wilcoxon signed-rank test) and a 10.9% RMSE reduction relative to the best classical result. The improvement is attributable to two factors: pot contamination correction ( $\text{RMSE } 0.264 \rightarrow 0.236$  kg) and the DINOv2 self-supervised ViT representation ( $\text{RMSE } 0.236 \rightarrow 0.213$  kg). The result validates the central hypothesis of this disserta-

tion: self-supervised Vision Transformer features provide richer, more biomass-discriminative representations than either classical geometric features or task-agnostic neural voxel embeddings.

#### **6.4.4 Research Question 4: Does increased angular sampling improve reconstruction quality?**

Yes, substantially. The 30-degree angular dual-camera protocol reduces mean ICP RMSE from 11.96 mm to 4.83 mm (59.6 % reduction) and increases merged point count by 188 %. When classical pipeline regression is trained on 30-degree protocol features, RMSE improves by 14.3 %. The 30-degree dataset therefore provides an independent accuracy improvement complementary to the proposed algorithmic advances.

### **6.5 RECOMMENDATIONS FOR FUTURE WORK**

#### **6.5.1 Domain-Specific Fine-Tuning of SAM 3D Objects**

The current implementation uses SAM 3D Objects in zero-shot inference mode. The systematic volume underestimation of 8.2 % relative to the classical pipeline (attributable to solid-object priors from Objaverse-XL training) could be reduced by fine-tuning the Geometry Model on a plant-specific 3-D dataset. The 30-degree dual-camera dataset collected in this study, combined with classical pipeline mesh reconstructions as pseudo-ground-truth 3-D shapes, could serve as a suitable fine-tuning corpus.

#### **6.5.2 Multi-Modal Sensor Fusion**

The present study uses RGB images as the sole SAM 3D Objects input, and depth images only for above-ground masking. Incorporating the calibrated Kinect V2 depth map as an additional input channel to the SAM 3D Objects Geometry Model would provide direct 3-D geometric constraints alongside DINOv2 features, potentially eliminating the volume underestimation bias and reducing sensitivity to single-view occlusions.

#### **6.5.3 Temporal Biomass Monitoring**

Extension to non-destructive, time-series biomass monitoring is a high-priority application. The single-image SAM 3D Objects inference capability is well-suited to temporal monitoring, requiring only one RGB photograph per monitoring time point without a turntable or multi-camera setup. Correspondence between estimated above-ground fresh weight and the dry biomass proxy must be established for time-series deployment.



### 6.5.4 Expansion to Additional Species and Field Environments

Validation on additional species (crops such as maize, soybean, and wheat, as well as indigenous tree species) and field environments (unstructured lighting, wind, partial occlusion) is required before the pipeline can claim broad generalisation. Field deployment would require replacing the turntable-based multi-view acquisition with a handheld or UAV-mounted sensor and replacing the depth-threshold mask with a learning-based ground plane segmentation approach.

### 6.5.5 Task-Aligned DeepVoxels Training

As discussed in Section 5.4, the DeepVoxels baseline’s underperformance is primarily attributable to training-objective misalignment. Training a shared DeepVoxels model across all 40 plants simultaneously, with biomass as an auxiliary regression head jointly optimised with the view-synthesis loss, could enable the neural voxel approach to match or exceed the Classical ANN without explicit geometric feature engineering.

### 6.5.6 Integration with Precision Agriculture Platforms

The proposed pipeline’s cost-performance profile (Kinect V2 at ~USD 150, single-image inference at 3.2 s per plant,  $R^2 = 0.834$ ) positions it as a viable candidate for commercial greenhouse integration. Future work should investigate API integration with existing nutrient management and yield prediction software, and the development of a mobile application that returns a biomass estimate via cloud inference of SAM 3D Objects from a single smartphone photograph.

## 6.6 CLOSING REMARKS

The results of this dissertation establish that self-supervised Vision Transformer features, operationalised through the DINOv2 backbone of Meta AI’s SAM 3D Objects foundation model [20, 18], provide a statistically significant and practically meaningful improvement over classical geometric features and learned neural voxel embeddings for automated above-ground biomass estimation of ornamental tree saplings. The combination of a lightweight depth-based pot isolation mask and single-image 3-D reconstruction via a self-supervised ViT encoder achieves  $R^2 = 0.834$  and  $RMSE = 0.213$  kg on a 40-plant LOOCV benchmark, representing the highest accuracy reported for this task on this sensor modality.

More broadly, these results demonstrate that general-purpose self-supervised pre-training on large, unlabelled image collections can produce visual representations that transfer effectively to the specialised domain of plant morphological analysis without task-specific labelling, substantially reducing the

data annotation burden that has historically constrained the application of deep learning to precision agriculture. As self-supervised foundation models continue to improve in scale and capability, their application to plant phenotyping—including biomass estimation, disease detection, yield prediction, and growth monitoring—is expected to become a central paradigm of computational agriculture research.

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## APPENDIX A DERIVATION OF HIGHER ORDER MODELS

## A.1 EXAMPLE HEADING

### A.1.1 Subheading example

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